

Research Figures and Tables

Evaluating Recommender Systems under Sparse User Feedback

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1 Experimental Pipeline

Table 1: Summary of the implemented experimental pipeline.

	Stage	Implementation
1	Raw data	MovieLens-1M and Amazon Reviews 2023 Video Games.
2	Positive implicit feedback	Ratings ≥ 4 retained as positive interactions.
3	Duplicate handling	Amazon repeated positive user–item pairs deduplicated by keeping the most recent positive interaction. MovieLens checked for positive user–item duplicates.
4	Core filtering	Iterative 5-core filtering applied to both users and items.
5	Temporal ordering	Timestamp used as primary ordering field; original source-row order breaks timestamp ties; deterministic <code>sequence_order</code> assigned.
6	Temporal split	Per-user leave-one-out: final interaction $\rightarrow$ test, second-last $\rightarrow$ validation, earlier interactions $\rightarrow$ training.
7	Sparsity simulation	Training histories reduced to nominal 50%, 25%, and 10% using random per-user, recent-history, and early-profile retention.
8	Catalogue control	Full post-preprocessing item catalogue kept fixed across all experimental conditions.
9	Hyperparameter tuning	Performed on baseline only using validation NDCG@10.
10	Final training	Selected baseline configuration frozen and reused across sparse conditions.
11	Evaluation	RecBole full-sort evaluation using Recall, HitRate, NDCG and MRR at $K \in \{5, 10, 20\}$.
12	Analysis	Absolute performance, relative change, rankings, seed variability, unseen targets and cross-dataset robustness.

2 Dataset and Preprocessing Tables

2.1 Raw dataset statistics

Table 2: Raw dataset statistics before positive-feedback conversion.

Dataset	Users	Items	Interactions / Reviews
MovieLens-1M	6,040	3,706	1,000,209
Amazon Video Games	2,766,656	137,249	4,624,615

2.2 Preprocessing progression

Table 3: MovieLens-1M preprocessing progression.

Stage	Users	Items	Interactions
Raw ratings	6,040	3,706	1,000,209
Positive implicit feedback ($r \geq 4$)	6,038	3,533	575,281
Iterative 5-core	6,034	3,125	574,376

Table 4: Amazon Video Games preprocessing progression.

Stage	Users	Items	Interactions
Raw reviews	2,766,656	137,249	4,624,615
Positive reviews before deduplication	2,115,867	120,360	3,445,132
Positive implicit feedback after deduplication	2,115,867	120,360	3,394,554
Iterative 5-core	63,413	19,022	533,550

Table 5: Additional preprocessing details.

Property	MovieLens-1M	Amazon Video Games
Positive threshold	Rating ≥ 4	Rating ≥ 4
Item identifier	Movie ID	<code>parent_asin</code>
Repeated positive pairs	Checked; none retained	50,578 repeated positive rows removed
Deduplication rule	Not required	Keep most recent positive user-item interaction
5-core iterations	2	11
Temporal field	<code>sequence_order</code>	<code>sequence_order</code>
Primary order	Timestamp	Timestamp
Tie-breaker	Original source-row order	Original source-row order

2.3 Final dataset and split statistics

Table 6: Final post-preprocessing dataset statistics.

Statistic	MovieLens-1M	Amazon Video Games
Users	6,034	63,413
Items in fixed catalogue	3,125	19,022
Total interactions	574,376	533,550
Training interactions	562,308	406,724
Validation interactions	6,034	63,413
Test interactions	6,034	63,413

2.4 User-history statistics

Table 7: Post-5-core user-history lengths before leave-one-out removal.

Statistic	MovieLens-1M	Amazon Video Games
Mean	95.19	8.41
Median	58	6
Minimum	5	5
Maximum	1,417	314

Table 8: Baseline training-history characteristics after validation and test targets are held out.

Statistic	MovieLens-1M	Amazon Video Games
Mean training interactions/user	93.19	6.41
Maximum sequential input length	200	50

3 Sparsity Design

3.1 Sparsity scenarios

Table 9: Implemented sparsity scenarios.

Repository label	Reported name	Retention rule	Interpretation
global	Random per-user	Random subset of each user’s training history	Controls interaction quantity while retaining interactions distributed across the user’s history.
recent	Recent-history	Most recent training interactions	Tests whether recent preference information is sufficient.
early	Early-profile	Earliest training interactions	Cold-start-like condition with known users; also introduces temporal staleness relative to the test target.

For every user,

$$n_{\text{retain}} = \max(1, \lceil n_{\text{training}} r \rceil). \quad (1)$$

For random per-user sparsity, seed 2025 is used and retained histories are nested:

$$10\% \subseteq 25\% \subseteq 50\% \subseteq 100\%. \quad (2)$$

3.2 Nominal and actual retention

Table 10: Nominal versus actual aggregate training retention.

Dataset	Nominal retention	Actual retention	Mean train interactions/user
MovieLens	100%	100.000%	93.19
MovieLens	50%	50.270%	46.85
MovieLens	25%	25.404%	23.67
MovieLens	10%	10.479%	9.76
Amazon	100%	100.000%	6.41
Amazon	50%	54.712%	3.51
Amazon	25%	29.767%	1.91
Amazon	10%	18.380%	1.18

3.3 MovieLens sparsity dataset statistics

Table 11: MovieLens training data and item exposure by sparsity condition.

Condition	Actual retention	Train interactions	Mean/user	Training-exposed items	Unseen test targets
Baseline	100.000%	562,308	93.19	3,125	0.000%
Random 50%	50.270%	282,674	46.85	3,122	0.033%
Random 25%	25.404%	142,846	23.67	3,065	0.265%
Random 10%	10.479%	58,922	9.76	2,853	1.425%
Recent 50%	50.270%	282,674	46.85	3,124	0.000%
Recent 25%	25.404%	142,846	23.67	3,102	0.066%
Recent 10%	10.479%	58,922	9.76	2,971	0.580%
Early 50%	50.270%	282,674	46.85	3,105	0.099%
Early 25%	25.404%	142,846	23.67	2,986	0.563%
Early 10%	10.479%	58,922	9.76	2,693	2.320%

3.4 Amazon sparsity dataset statistics

Table 12: Amazon training data and item exposure by sparsity condition.

Condition	Actual retention	Train interactions	Mean/user	Training-exposed items	Unseen test targets
Baseline	100.000%	406,724	6.41	18,966	0.358%
Random 50%	54.712%	222,525	3.51	18,565	1.667%
Random 25%	29.767%	121,068	1.91	17,109	4.889%
Random 10%	18.380%	74,756	1.18	15,029	9.071%
Recent 50%	54.712%	222,525	3.51	18,604	1.116%
Recent 25%	29.767%	121,068	1.91	17,075	3.301%
Recent 10%	18.380%	74,756	1.18	14,919	6.894%
Early 50%	54.712%	222,525	3.51	18,184	3.331%
Early 25%	29.767%	121,068	1.91	16,375	8.621%
Early 10%	18.380%	74,756	1.18	14,478	13.177%

4 Models and Training Configuration

4.1 Evaluated models

Table 13: Evaluated recommender systems and model families.

Model	Family	Final runs per condition
Pop	Non-personalized	1
ItemKNN	Classical	1
BPR	Classical	3
EASE	Linear	1
NeuMF	Neural	3
MultiVAE	Neural	3
GRU4Rec	Sequential	3
SASRec	Sequential	3
BERT4Rec	Sequential	3
LightGCN	Graph	3

Stochastic models use seeds 2025, 2026 and 2027.

4.2 Selected model configurations

Table 14: Important selected hyperparameters used in the final experiments.

Model	MovieLens-1M	Amazon Video Games
Pop	No model hyperparameters	No model hyperparameters
ItemKNN	$k = 50$, shrink = 0	$k = 200$, shrink = 0
BPR	Embedding = 64, weight decay = 10^{-6}	Embedding = 64, weight decay = 10^{-6}
EASE	Regularization = 500	Regularization = 100
NeuMF	MF = 64, MLP = 32, layers [64, 32, 16, 8]	MF = 64, MLP = 64, layers [64, 32, 16, 8]
MultiVAE	Latent = 200, hidden = [600], dropout = 0.5	Latent = 200, hidden = [600], dropout = 0.5
GRU4Rec	Embedding = 64, hidden = 128, layers = 1, dropout = 0.2	Embedding = 64, hidden = 64, layers = 1, dropout = 0.2
SASRec	Hidden = 64, layers = 2, heads = 2, dropout = 0.1	Hidden = 64, layers = 1, heads = 2, dropout = 0.2
BERT4Rec	Hidden = 64, layers = 2, heads = 2, mask = 0.2	Hidden = 64, layers = 2, heads = 2, mask = 0.2
LightGCN	Embedding = 64, layers = 1, reg = 10^{-5}	Embedding = 64, layers = 2, reg = 10^{-5}

4.3 Common experimental settings

Table 15: Common training and evaluation settings.

Setting	Value
Framework	RecBole 1.2.1
Primary validation metric	NDCG@10
Temporal field	sequence_order
Evaluation split	Temporal leave-one-out
Evaluation mode	Full-sort
User grouping	Enabled
Candidate catalogue	Fixed post-preprocessing catalogue
MovieLens sequence limit	200
Amazon sequence limit	50
Random sparsity seed	2025
Stochastic model seeds	2025, 2026, 2027
Hyperparameter tuning	Baseline only; configuration frozen for sparsity experiments

5 Evaluation Metrics

Table 16: Evaluation metrics used in the study.

Metric	Interpretation	Calculation in leave-one-out setting
Recall@K	Whether the relevant test item is retrieved within the top K .	With one relevant test item per user, Recall@K is either 0 or 1 for that user.
HitRate@K	Whether at least one relevant item occurs in the top K .	Equivalent to Recall@K because there is exactly one relevant test target per user.
NDCG@K	Rewards retrieving the relevant item and placing it near the top of the ranking.	For relevant-item rank $r \leq K$, contribution is proportional to $1/\log_2(r+1)$; otherwise 0.
MRR@K	Measures the reciprocal rank of the relevant test item.	For rank $r \leq K$, contribution is $1/r$; otherwise 0.

Metrics are reported at

$$K \in \{5, 10, 20\}. \quad (3)$$

Therefore, the recorded metric set is:

Table 17: Complete metric set.

Recall	HitRate	NDCG	MRR
Recall@5	Hit@5	NDCG@5	MRR@5
Recall@10	Hit@10	NDCG@10	MRR@10
Recall@20	Hit@20	NDCG@20	MRR@20

NDCG@10 is the primary analysis metric.

6 Baseline Results Tables

6.1 MovieLens baseline metrics

Table 18: MovieLens-1M baseline results. Means and standard deviations are shown for stochastic models.

Model	Family	NDCG@10	SD	Recall@10	SD	MRR@10	SD
SASRec	Sequential	0.119079	0.002283	0.223898	0.001195	0.087307	0.003045
GRU4Rec	Sequential	0.112553	0.001456	0.210032	0.004928	0.083042	0.001652
BERT4Rec	Sequential	0.106708	0.002586	0.212463	0.002713	0.074836	0.002591
EASE	Linear	0.045156	–	0.090322	–	0.031692	–
MultiVAE	Neural	0.039195	0.000607	0.081151	0.001066	0.026741	0.000433
LightGCN	Graph	0.038068	0.000704	0.079604	0.002226	0.025681	0.000365
BPR	Classical	0.038047	0.001206	0.079218	0.002738	0.025787	0.000800
ItemKNN	Classical	0.037286	–	0.075572	–	0.025903	–
NeuMF	Neural	0.031950	0.000887	0.067120	0.001868	0.021448	0.000636
Pop	Non-personalized	0.019158	–	0.039443	–	0.013127	–

6.2 Amazon baseline metrics

Table 19: Amazon Video Games baseline results. Means and standard deviations are shown for stochastic models.

Model	Family	NDCG@10	SD	Recall@10	SD	MRR@10	SD
SASRec	Sequential	0.046315	0.000368	0.096300	0.001948	0.031080	0.001087
GRU4Rec	Sequential	0.044938	0.000974	0.084983	0.001570	0.032841	0.000800
EASE	Linear	0.040555	–	0.072666	–	0.030816	–
BERT4Rec	Sequential	0.039125	0.000320	0.074291	0.000951	0.028502	0.000269
LightGCN	Graph	0.035857	0.000431	0.065933	0.000523	0.026761	0.000412
BPR	Classical	0.035450	0.000467	0.065002	0.001148	0.026504	0.000320
MultiVAE	Neural	0.034618	0.000439	0.064361	0.000593	0.025627	0.000393
NeuMF	Neural	0.024331	0.000272	0.046247	0.000587	0.017706	0.000180
ItemKNN	Classical	0.024051	–	0.043934	–	0.018025	–
Pop	Non-personalized	0.016255	–	0.032012	–	0.011597	–

6.3 NDCG across recommendation-list lengths

Table 20: MovieLens baseline NDCG at $K = 5, 10, 20$.

Model	NDCG@5	NDCG@10	NDCG@20
SASRec	0.093748	0.119079	0.145713
GRU4Rec	0.088337	0.112553	0.137308
BERT4Rec	0.079638	0.106708	0.133973
EASE	0.032601	0.045156	0.061191
MultiVAE	0.027150	0.039195	0.053790
LightGCN	0.026700	0.038068	0.052614
BPR	0.026360	0.038047	0.052876
ItemKNN	0.026617	0.037286	0.050919
NeuMF	0.022227	0.031950	0.045409
Pop	0.013566	0.019158	0.028064

Table 21: Amazon baseline NDCG at $K = 5, 10, 20$.

Model	NDCG@5	NDCG@10	NDCG@20
SASRec	0.034908	0.046315	0.057919
GRU4Rec	0.034953	0.044938	0.055887
EASE	0.032719	0.040555	0.048094
BERT4Rec	0.030387	0.039125	0.048845
LightGCN	0.028336	0.035857	0.043960
BPR	0.028151	0.035450	0.043304
MultiVAE	0.027229	0.034618	0.042207
NeuMF	0.018951	0.024331	0.030441
ItemKNN	0.019178	0.024051	0.029061
Pop	0.011771	0.016255	0.019940

6.4 Baseline seed variability

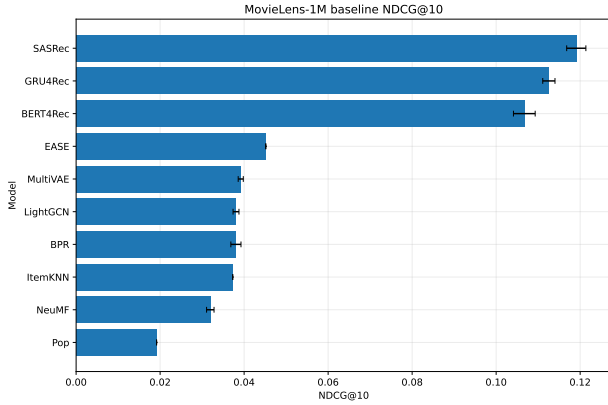
Table 22: MovieLens baseline NDCG@10 seed variability for stochastic models.

Model	Mean	SD	Range	CV
BERT4Rec	0.106708	0.002586	0.005143	2.42%
SASRec	0.119079	0.002283	0.004005	1.92%
GRU4Rec	0.112553	0.001456	0.002551	1.29%
BPR	0.038047	0.001206	0.002410	3.17%
NeuMF	0.031950	0.000887	0.001656	2.77%
LightGCN	0.038068	0.000704	0.001407	1.85%
MultiVAE	0.039195	0.000607	0.001093	1.55%

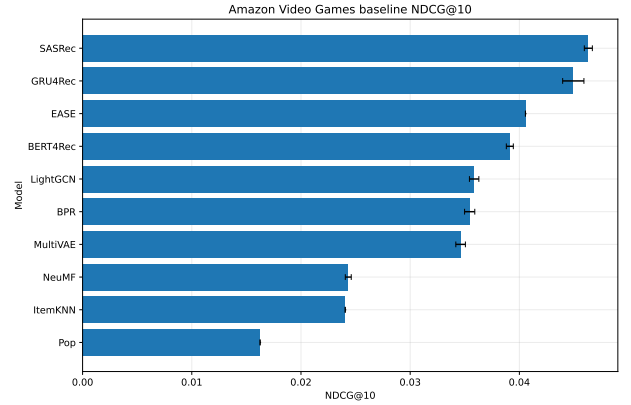
Table 23: Amazon baseline NDCG@10 seed variability for stochastic models.

Model	Mean	SD	Range	CV
GRU4Rec	0.044938	0.000974	0.001822	2.17%
BPR	0.035450	0.000467	0.000924	1.32%
MultiVAE	0.034618	0.000439	0.000877	1.27%
LightGCN	0.035857	0.000431	0.000856	1.20%
SASRec	0.046315	0.000368	0.000698	0.79%
BERT4Rec	0.039125	0.000320	0.000639	0.82%
NeuMF	0.024331	0.000272	0.000493	1.12%

7 Baseline Figures

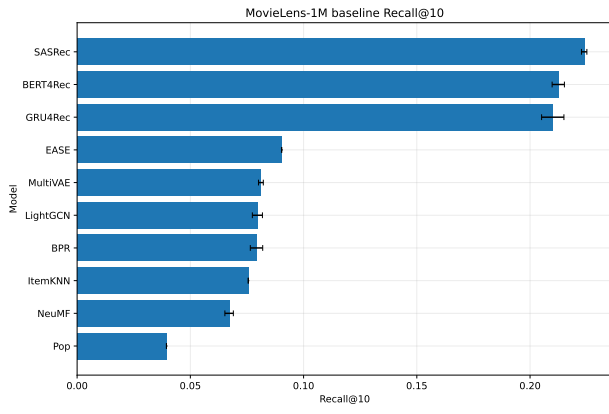


(a) MovieLens-1M.

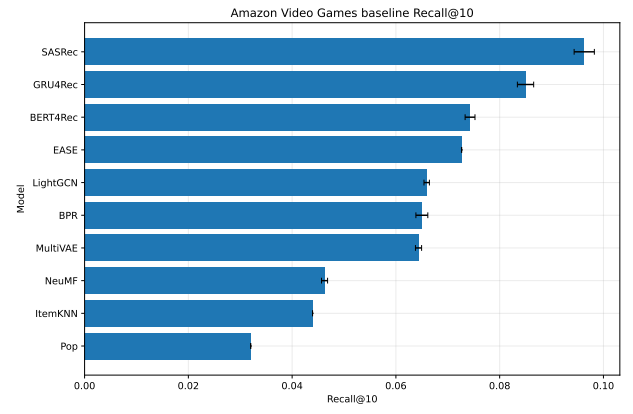


(b) Amazon Video Games.

Figure 1: Baseline NDCG@10 performance across all models. Error bars indicate one sample standard deviation for stochastic models.

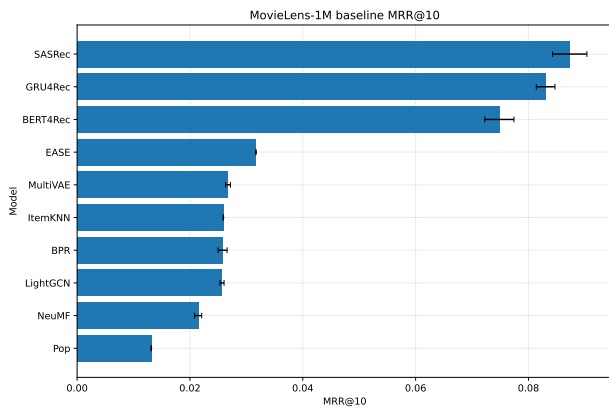


(a) MovieLens-1M.

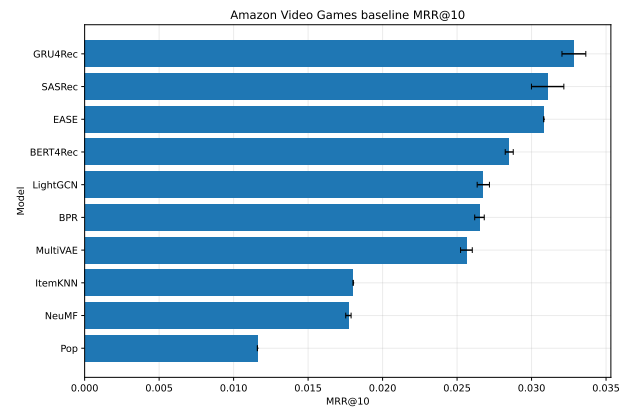


(b) Amazon Video Games.

Figure 2: Baseline Recall@10 performance.



(a) MovieLens-1M.



(b) Amazon Video Games.

Figure 3: Baseline MRR@10 performance.

8 MovieLens Sparsity Figures

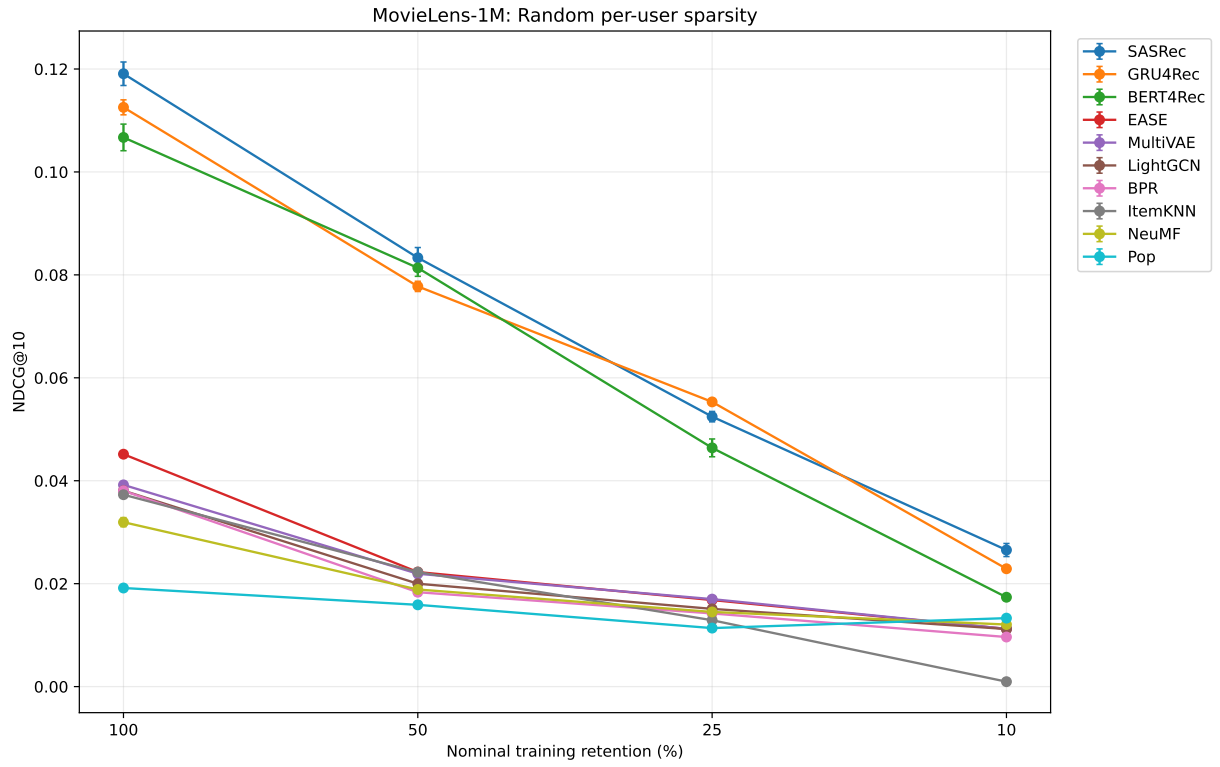


Figure 4: MovieLens NDCG@10 performance under random per-user sparsity as training-history retention decreases.

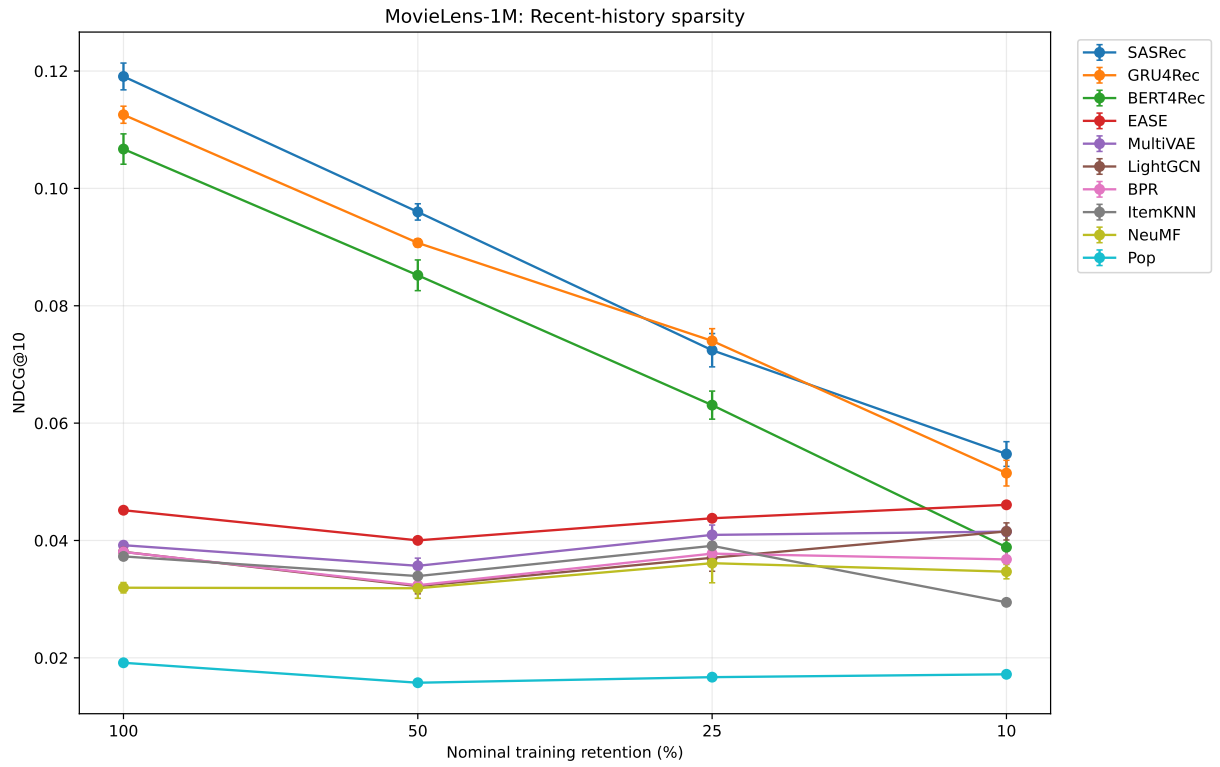


Figure 5: MovieLens NDCG@10 performance under recent-history sparsity as training-history retention decreases.

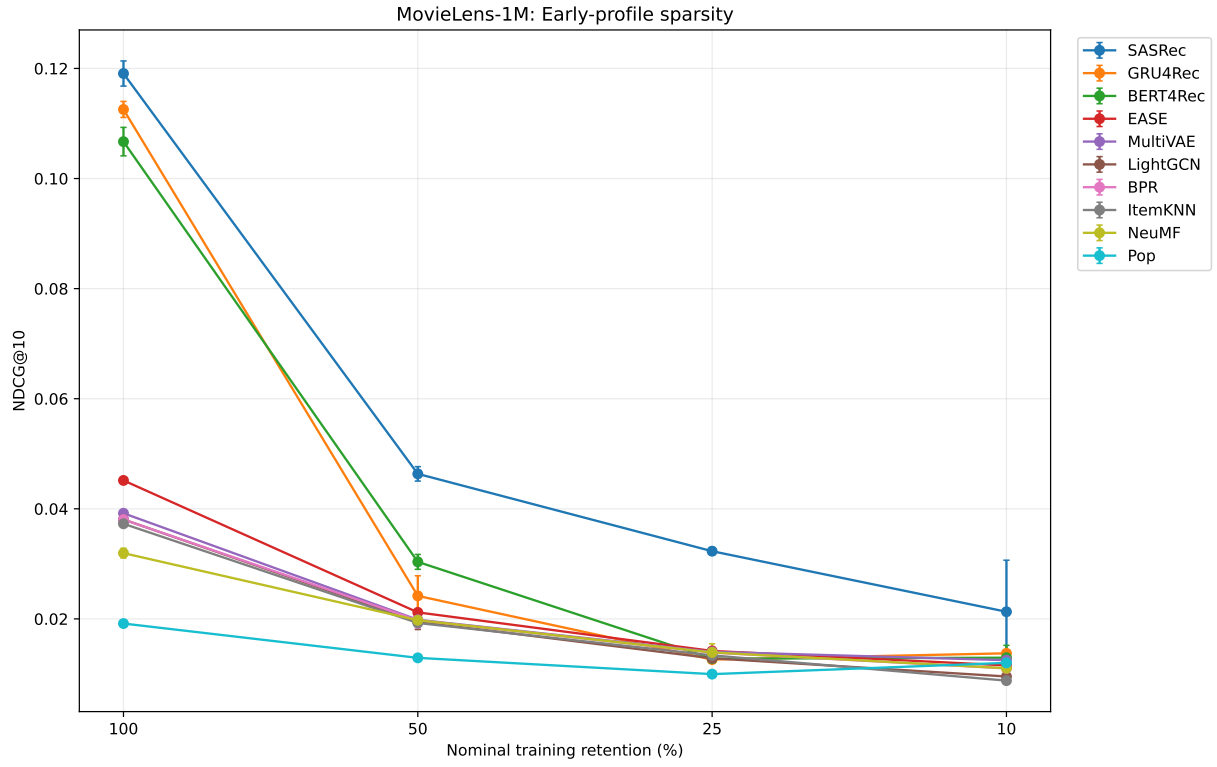


Figure 6: MovieLens NDCG@10 performance under early-profile sparsity as training-history retention decreases.

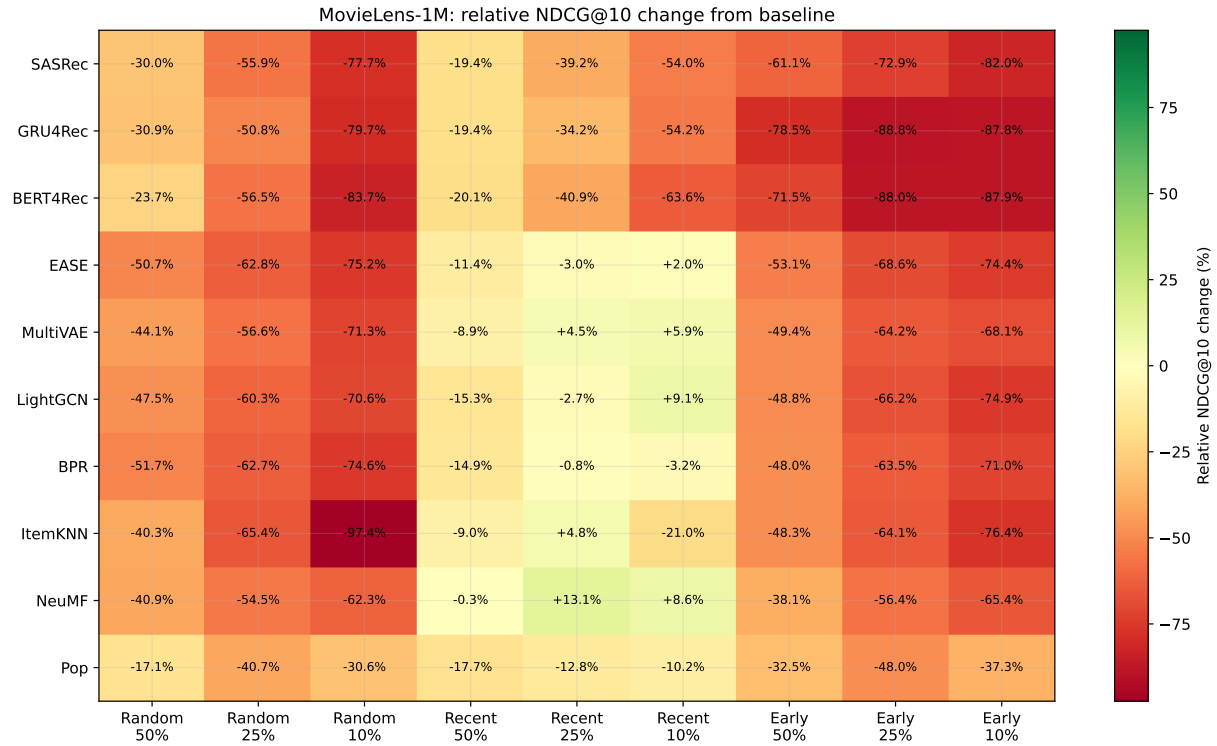


Figure 7: MovieLens relative NDCG@10 change from each model's full-data baseline. Negative values indicate degradation; positive values indicate improvement.

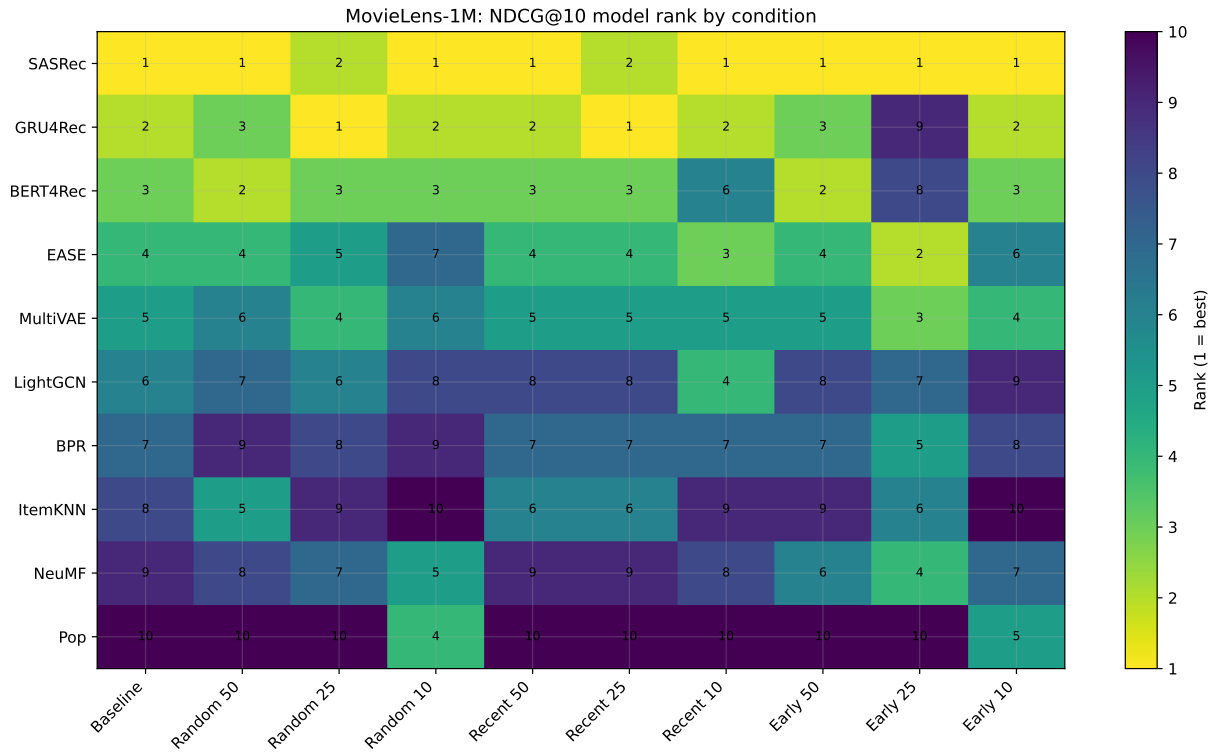


Figure 8: MovieLens NDCG@10 model ranks across baseline and sparsity conditions.

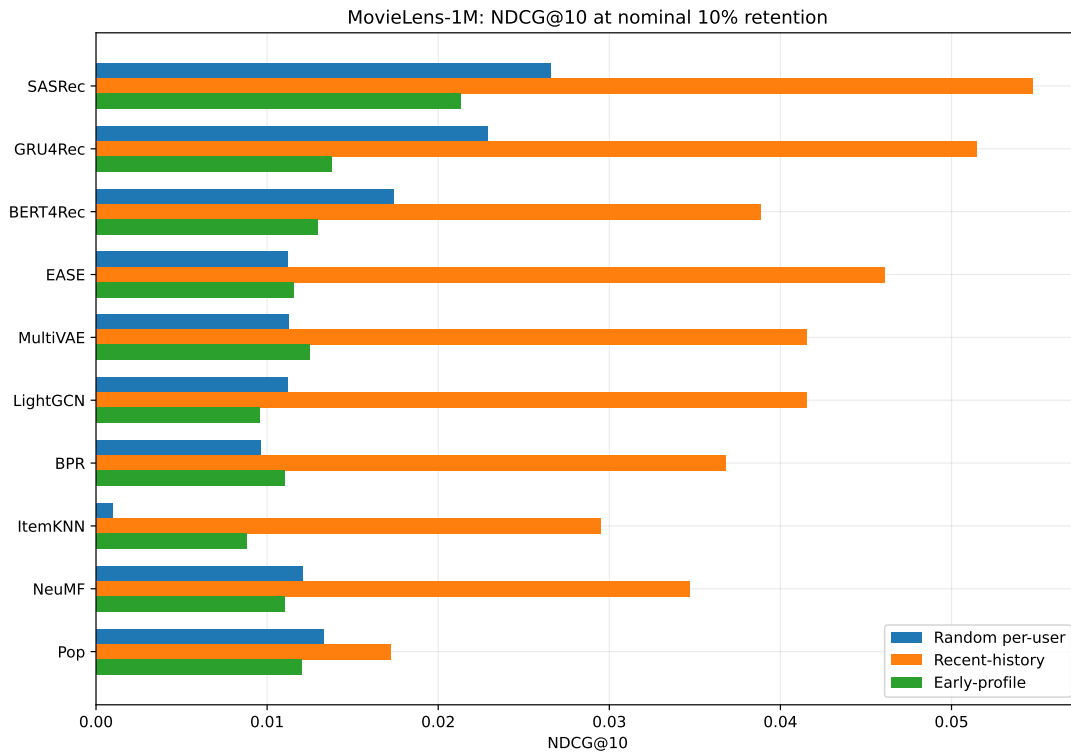


Figure 9: MovieLens NDCG@10 under the three nominal 10% sparsity scenarios.

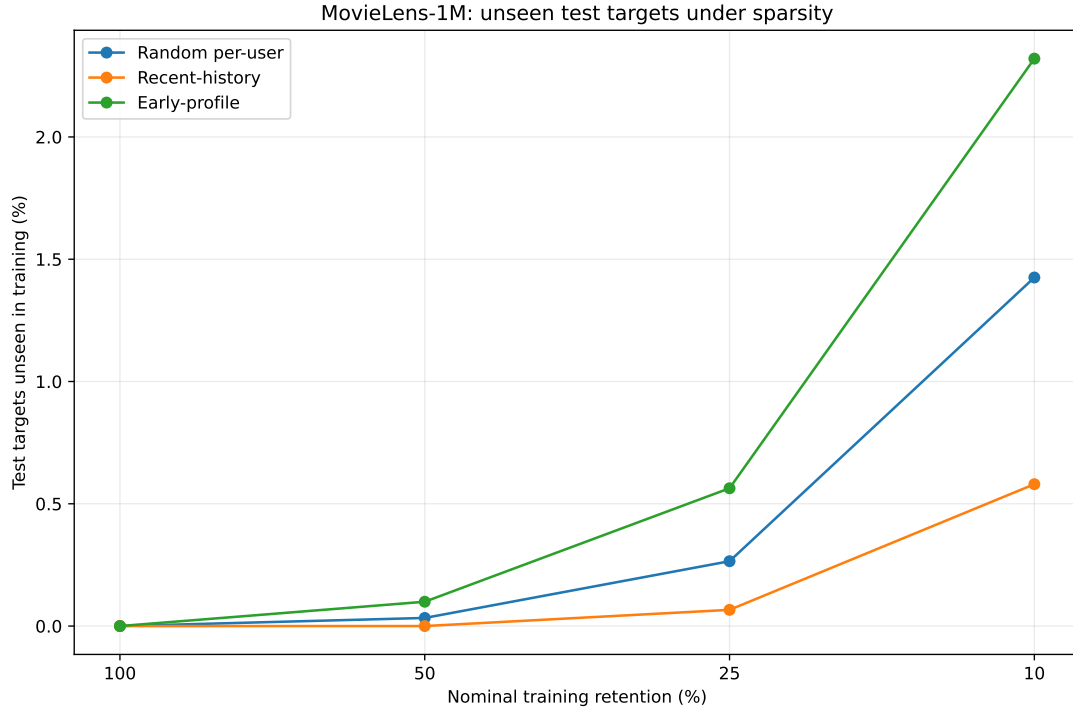


Figure 10: Percentage of MovieLens held-out test targets whose items become globally unseen in the retained training interactions.

9 Amazon Sparsity Figures

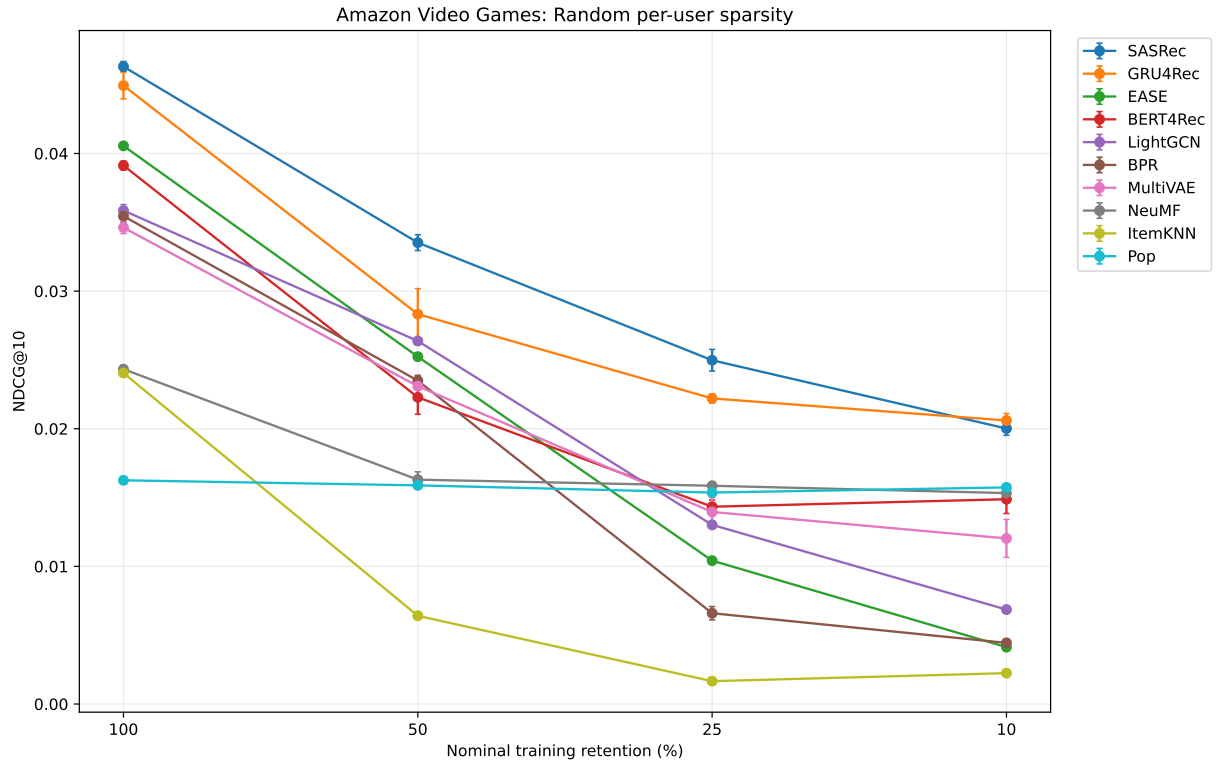


Figure 11: Amazon NDCG@10 performance under random per-user sparsity as training-history retention decreases.

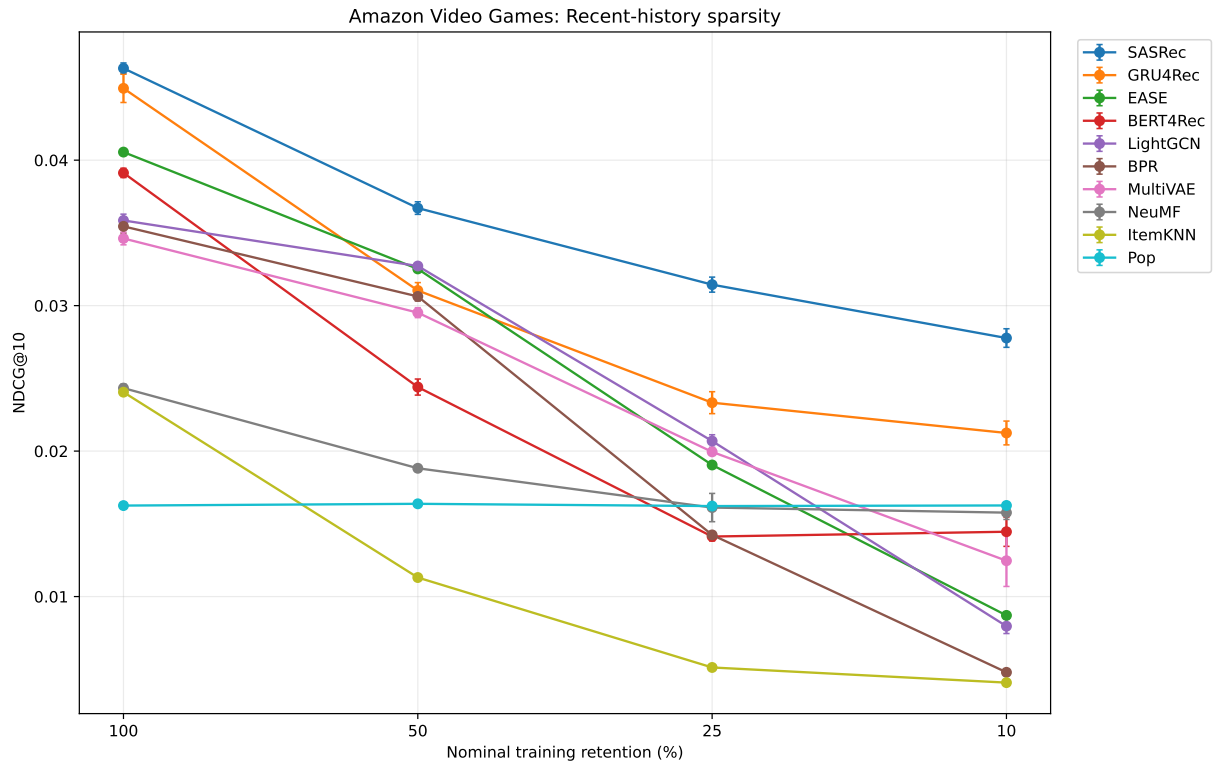


Figure 12: Amazon NDCG@10 performance under recent-history sparsity as training-history retention decreases.

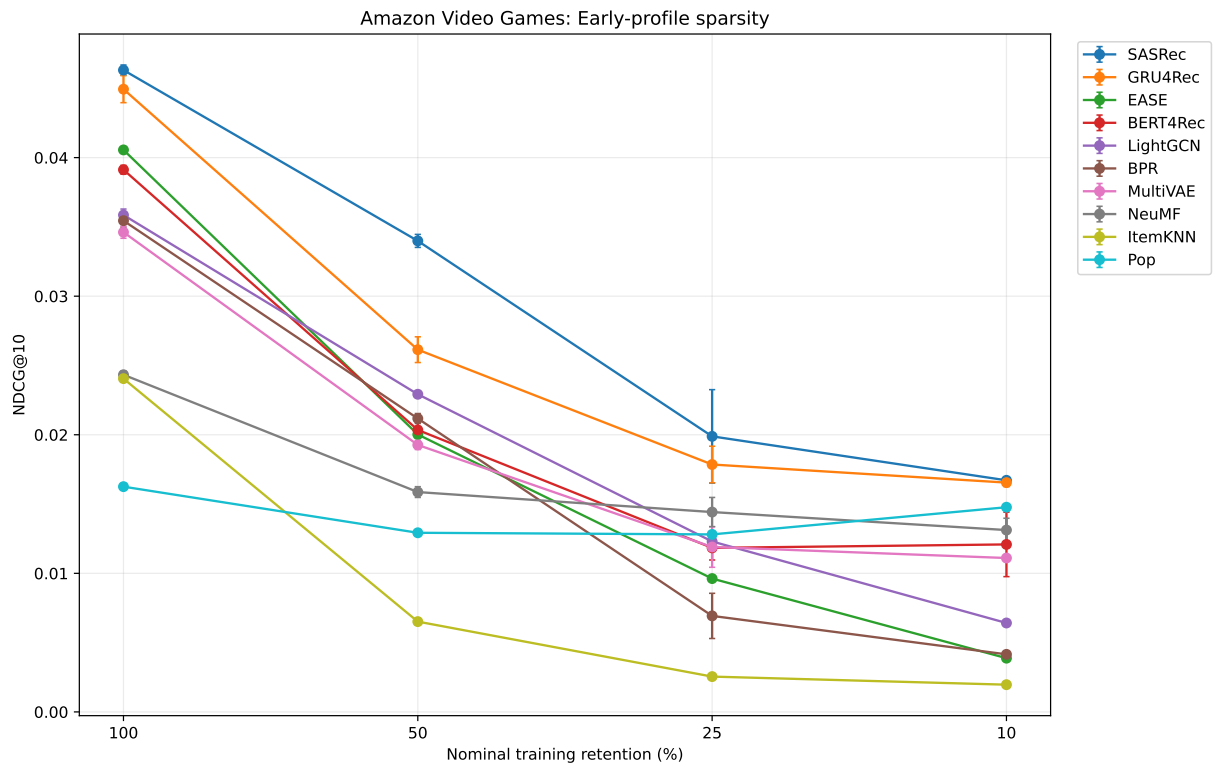


Figure 13: Amazon NDCG@10 performance under early-profile sparsity as training-history retention decreases.

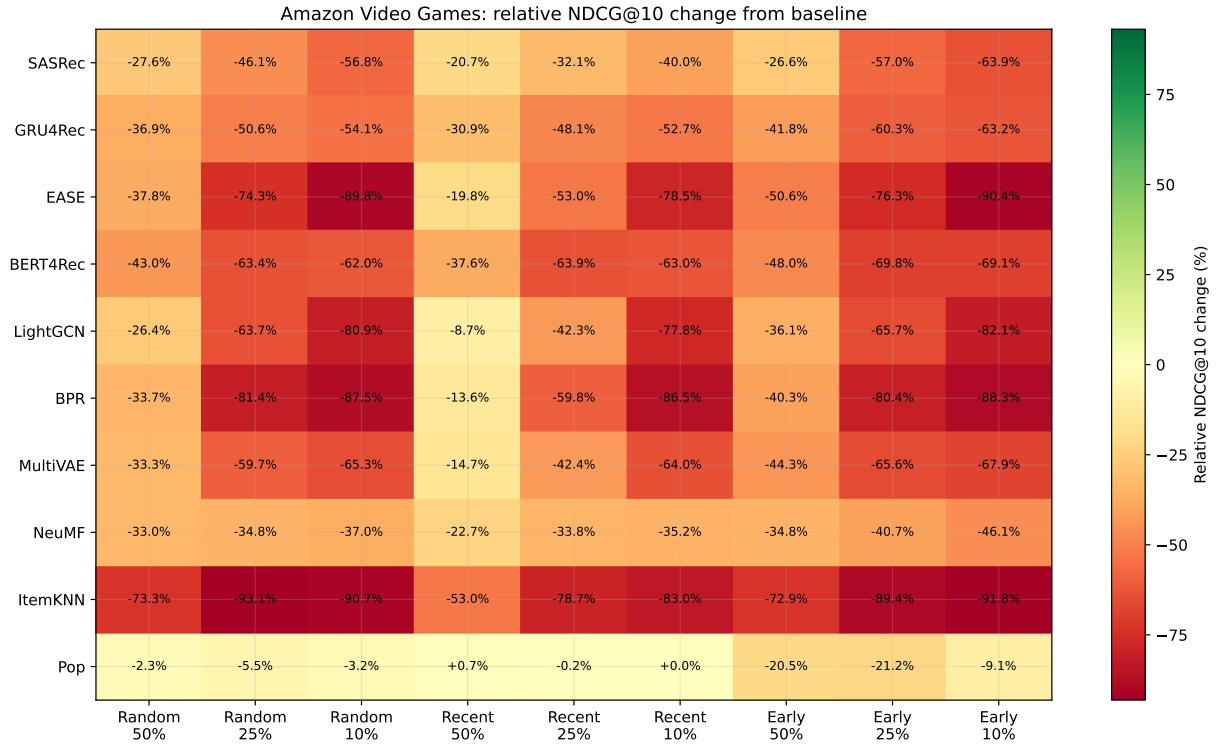


Figure 14: Amazon relative NDCG@10 change from each model’s full-data baseline. Negative values indicate degradation; positive values indicate improvement.

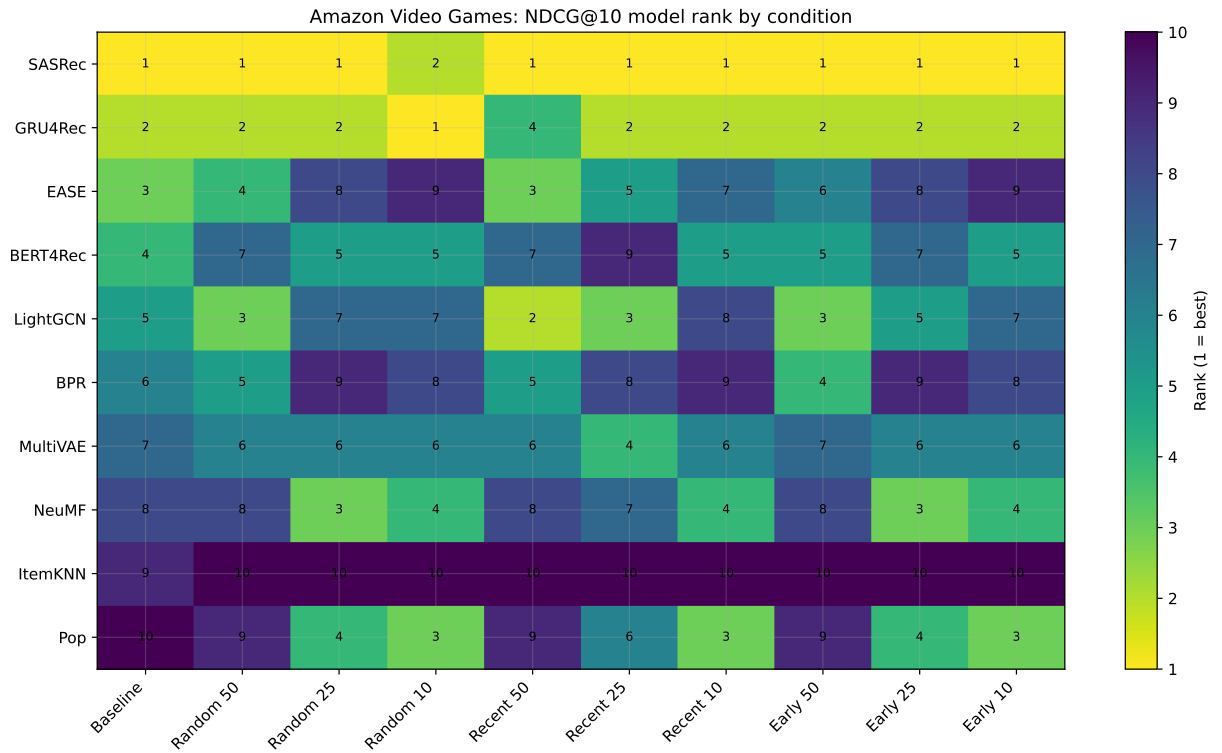


Figure 15: Amazon NDCG@10 model ranks across baseline and sparsity conditions.

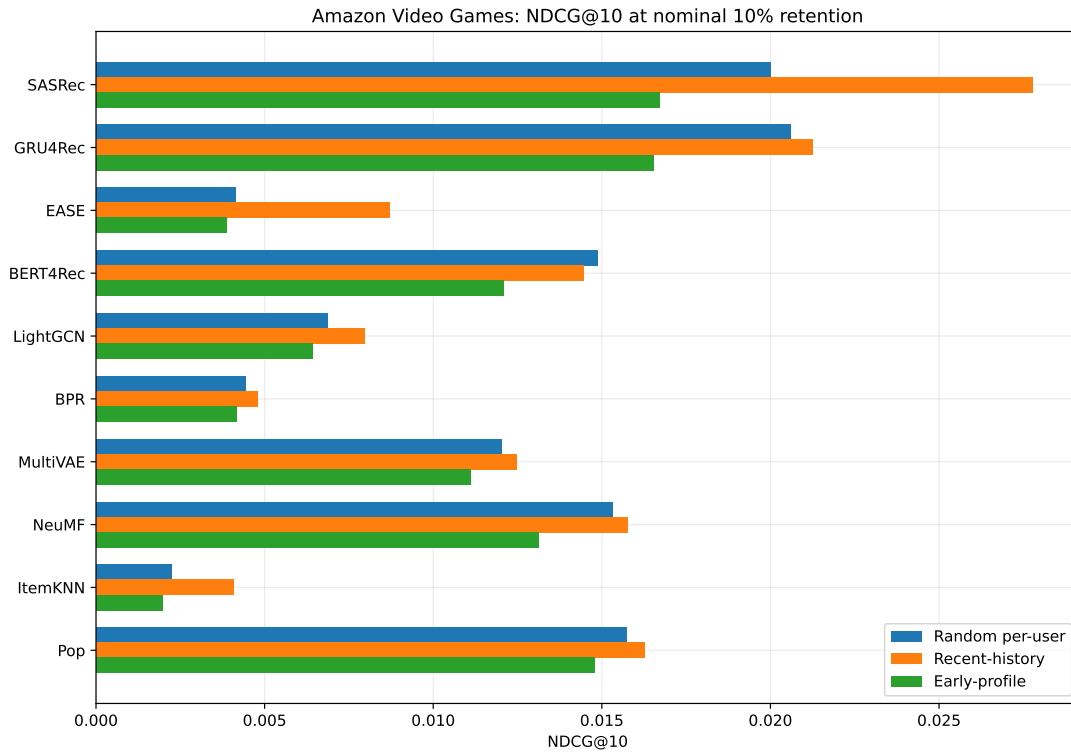


Figure 16: Amazon NDCG@10 under the three nominal 10% sparsity scenarios.

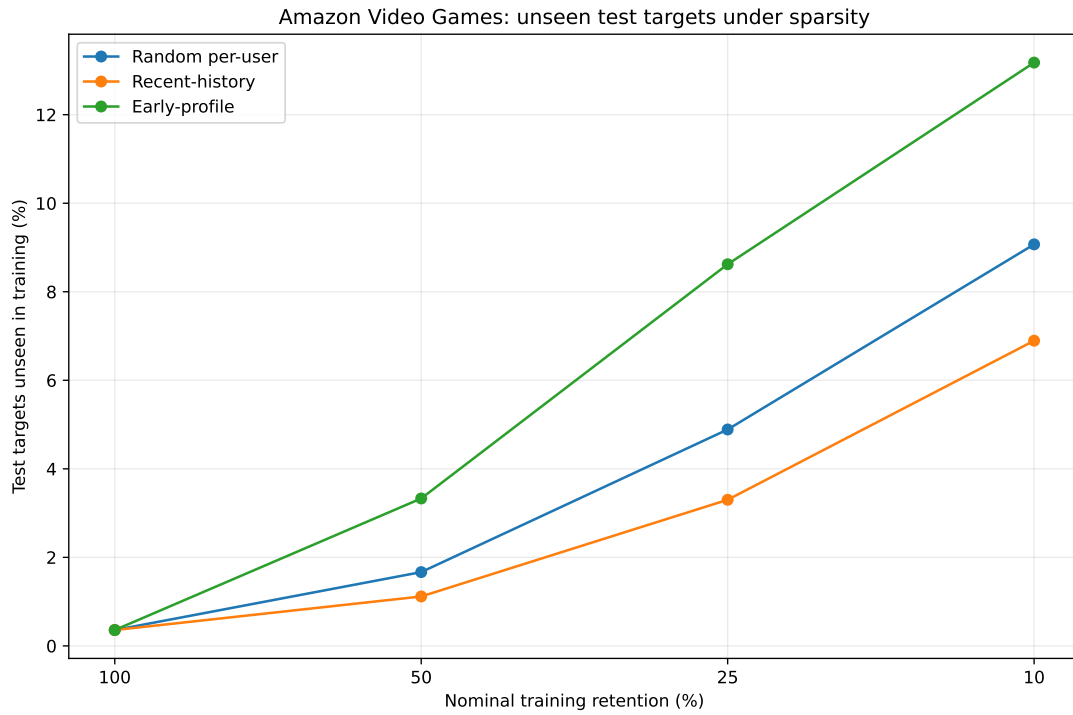


Figure 17: Percentage of Amazon held-out test targets whose items become globally unseen in the retained training interactions.

10 Severe Sparsity Tables

10.1 MovieLens nominal 10% NDCG@10

Table 24: MovieLens NDCG@10 at nominal 10% retention.

Model	Random	Recent	Early
SASRec	0.026539	0.054733	0.021299
GRU4Rec	0.022897	0.051480	0.013747
BERT4Rec	0.017363	0.038849	0.012949
EASE	0.011213	0.046078	0.011551
MultiVAE	0.011223	0.041512	0.012501
LightGCN	0.011183	0.041545	0.009538
BPR	0.009634	0.036775	0.011007
ItemKNN	0.000957	0.029457	0.008786
NeuMF	0.012062	0.034686	0.011035
Pop	0.013303	0.017196	0.012014

10.2 Amazon nominal 10% NDCG@10

Table 25: Amazon NDCG@10 at nominal 10% retention.

Model	Random	Recent	Early
SASRec	0.020016	0.027772	0.016713
GRU4Rec	0.020602	0.021246	0.016547
EASE	0.004134	0.008705	0.003878
BERT4Rec	0.014880	0.014456	0.012086
LightGCN	0.006863	0.007964	0.006417
BPR	0.004441	0.004792	0.004154
MultiVAE	0.012034	0.012462	0.011106
NeuMF	0.015325	0.015770	0.013123
ItemKNN	0.002242	0.004084	0.001965
Pop	0.015739	0.016263	0.014770

10.3 Relative change under severe sparsity

Table 26: Relative NDCG@10 change from baseline under nominal 10% retention. Negative values represent degradation and positive values improvement.

Model	MovieLens-1M			Amazon Video Games		
	Random	Recent	Early	Random	Recent	Early
Pop	-30.56%	-10.24%	-37.29%	-3.17%	+0.05%	-9.14%
ItemKNN	-97.43%	-21.00%	-76.44%	-90.68%	-83.02%	-91.83%
BPR	-74.64%	-3.21%	-71.02%	-87.46%	-86.48%	-88.28%
EASE	-75.17%	+2.04%	-74.42%	-89.81%	-78.54%	-90.44%
NeuMF	-62.31%	+8.56%	-65.39%	-36.99%	-35.17%	-46.08%
MultiVAE	-71.34%	+5.94%	-68.09%	-65.27%	-64.03%	-67.91%
GRU4Rec	-79.66%	-54.24%	-87.78%	-54.13%	-52.71%	-63.16%
SASRec	-77.71%	-54.03%	-82.01%	-56.79%	-40.03%	-63.91%
BERT4Rec	-83.73%	-63.56%	-87.88%	-61.96%	-63.05%	-69.12%
LightGCN	-70.60%	+9.13%	-74.94%	-80.86%	-77.78%	-82.10%

11 Sparse-Condition Seed Variability

Table 27: Largest MovieLens sparse-condition NDCG@10 coefficients of variation.

Scenario	Retention	Model	SD	CV
Early-profile	10%	SASRec	0.009363	43.96%
Early-profile	10%	BERT4Rec	0.002247	17.35%
Early-profile	50%	GRU4Rec	0.003654	15.11%
Random	10%	NeuMF	0.001589	13.18%
Random	25%	LightGCN	0.001894	12.54%
Early-profile	25%	NeuMF	0.001554	11.17%
Random	25%	BPR	0.001337	9.40%
Recent-history	25%	NeuMF	0.003340	9.24%

Table 28: Largest Amazon sparse-condition NDCG@10 coefficients of variation.

Scenario	Retention	Model	SD	CV
Early-profile	25%	BPR	0.001631	23.54%
Early-profile	10%	BERT4Rec	0.002325	19.23%
Early-profile	25%	SASRec	0.003371	16.95%
Recent-history	10%	MultiVAE	0.001768	14.18%
Early-profile	25%	MultiVAE	0.001465	12.31%
Random	10%	MultiVAE	0.001380	11.47%
Random	25%	MultiVAE	0.001044	7.48%
Early-profile	25%	BERT4Rec	0.000876	7.40%

12 Cross-Dataset Results Tables

12.1 Baseline rank comparison

Table 29: Baseline NDCG@10 model rankings across datasets.

Model	MovieLens rank	Amazon rank	Amazon – MovieLens
SASRec	1	1	0
GRU4Rec	2	2	0
BERT4Rec	3	4	+1
EASE	4	3	-1
MultiVAE	5	7	+2
LightGCN	6	5	-1
BPR	7	6	-1
ItemKNN	8	9	+1
NeuMF	9	8	-1
Pop	10	10	0

Baseline rank agreement:

$$\rho_{\text{Spearman}} = 0.939. \quad (4)$$

12.2 Median relative change

Table 30: Median relative NDCG@10 change across the ten models.

Scenario	Nominal retention	MovieLens	Amazon
Random	50%	-40.62%	-33.50%
Random	25%	-56.57%	-61.52%
Random	10%	-74.91%	-63.61%
Recent	50%	-15.10%	-20.27%
Recent	25%	-2.87%	-45.22%
Recent	10%	-6.73%	-63.54%
Early	50%	-49.08%	-41.02%
Early	25%	-65.24%	-65.64%
Early	10%	-74.68%	-68.52%

12.3 Cross-dataset robustness agreement at nominal 10%

Table 31: Spearman agreement between model relative robustness on MovieLens and Amazon at nominal 10% retention.

Scenario	Spearman ρ
Random per-user	0.467
Recent-history	-0.152
Early-profile	0.309

12.4 Rank agreement by condition

Table 32: MovieLens–Amazon NDCG@10 model-rank agreement.

Condition	Spearman ρ
Baseline	0.939
Random 50%	0.491
Random 25%	0.552
Random 10%	0.915
Recent 50%	0.515
Recent 25%	0.382
Recent 10%	0.370
Early 50%	0.648
Early 25%	-0.018
Early 10%	0.794

12.5 Unseen-target comparison

Table 33: Percentage of held-out test targets globally unseen in retained training interactions.

Scenario	Retention	MovieLens	Amazon
Baseline	100%	0.000%	0.358%
Random	50%	0.033%	1.667%
Random	25%	0.265%	4.889%
Random	10%	1.425%	9.071%
Recent	50%	0.000%	1.116%
Recent	25%	0.066%	3.301%
Recent	10%	0.580%	6.894%
Early	50%	0.099%	3.331%
Early	25%	0.563%	8.621%
Early	10%	2.320%	13.177%

12.6 Condition winners

Table 34: Highest mean NDCG@10 model in each experimental condition.

Condition	MovieLens-1M	Amazon Video Games
Baseline	SASRec	SASRec
Random 50%	SASRec	SASRec
Random 25%	GRU4Rec	SASRec
Random 10%	SASRec	GRU4Rec
Recent 50%	SASRec	SASRec
Recent 25%	GRU4Rec	SASRec
Recent 10%	SASRec	SASRec
Early 50%	SASRec	SASRec
Early 25%	SASRec	SASRec
Early 10%	SASRec	SASRec

13 Metric and K Consistency

13.1 Metric-ranking consistency

Table 35: Spearman agreement between NDCG@10 rankings and supporting metric rankings.

Condition	ML NDCG–Recall	ML NDCG–MRR	Amazon NDCG–Recall	Amazon NDCG–MRR
Baseline	0.988	0.952	0.988	0.976
Random 50%	0.964	0.988	0.988	1.000
Random 25%	0.976	1.000	0.988	1.000
Random 10%	0.988	0.952	0.988	1.000
Recent 50%	0.988	1.000	0.964	0.939
Recent 25%	0.988	1.000	0.976	0.988
Recent 10%	0.988	0.988	1.000	1.000
Early 50%	0.988	0.867	0.988	0.988
Early 25%	0.927	0.976	0.952	0.927
Early 10%	0.976	0.964	1.000	1.000

13.2 NDCG ranking stability across K

Table 36: Spearman agreement of NDCG rankings across recommendation-list lengths.

Condition	ML @5 vs @10	ML @10 vs @20	Amazon @5 vs @10	Amazon @10 vs @20
Baseline	0.988	0.988	0.976	0.988
Random 50%	0.988	1.000	1.000	1.000
Random 25%	0.988	0.988	1.000	1.000
Random 10%	0.952	0.988	0.964	1.000
Recent 50%	1.000	0.988	0.976	0.988
Recent 25%	1.000	0.988	0.988	0.988
Recent 10%	0.988	1.000	0.988	1.000
Early 50%	0.842	0.927	0.988	0.988
Early 25%	0.976	0.794	0.915	0.964
Early 10%	0.939	0.952	0.988	0.988

14 Cross-Dataset Figures

14.1 Baseline ranking

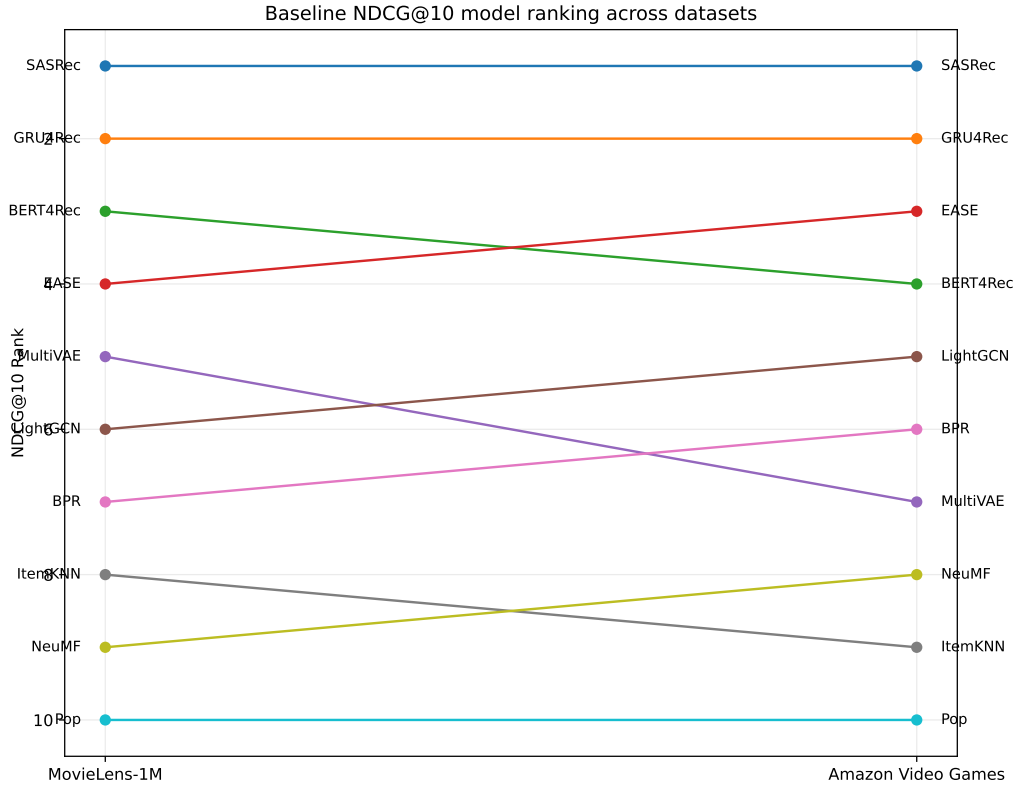


Figure 18: Baseline NDCG@10 model-rank movement between MovieLens-1M and Amazon Video Games.

14.2 Median relative performance change

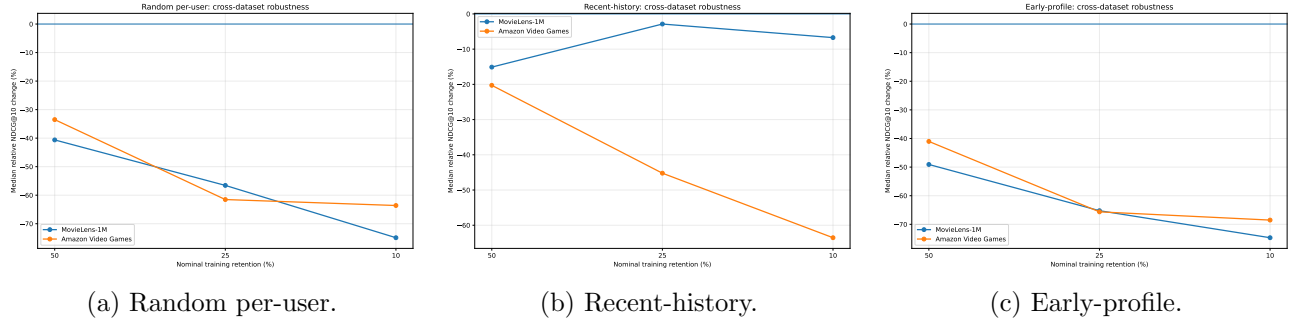


Figure 19: Median relative NDCG@10 change across the ten models on each dataset.

14.3 Model-level severe robustness

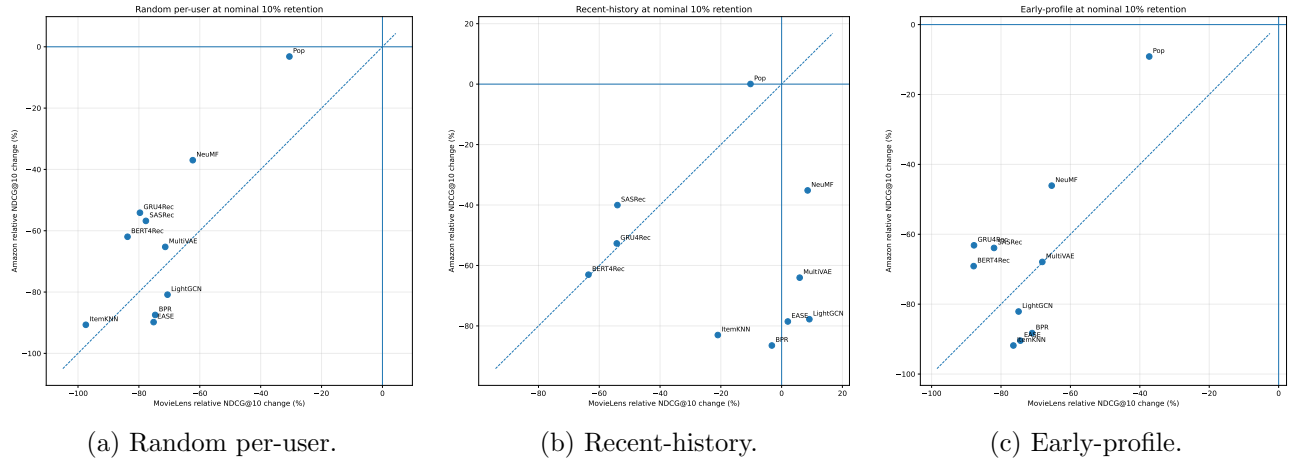


Figure 20: Model-level relative NDCG@10 change across datasets at nominal 10% retention. The diagonal represents equal relative change on the two datasets.

14.4 Cross-dataset rank agreement

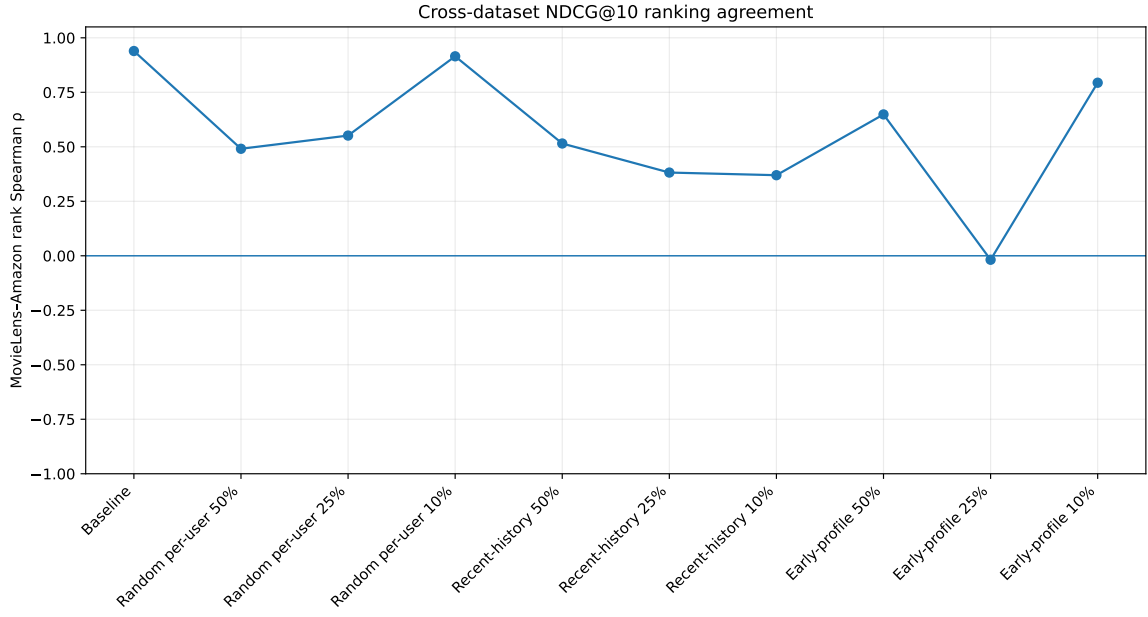


Figure 21: Spearman agreement between MovieLens and Amazon NDCG@10 model rankings across matching experimental conditions.

14.5 Cross-dataset unseen targets

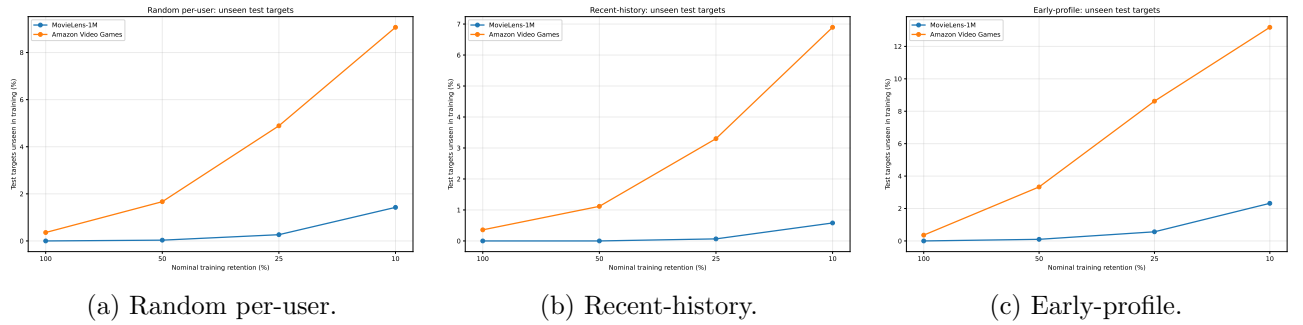


Figure 22: Percentage of held-out test targets whose items become globally unseen in retained training data.

15 Compact Key-Results Tables

Table 37: Compact summary of the main quantitative results.

Result	Value / observation
Best MovieLens baseline model	SASRec, NDCG@10 = 0.119079
Best Amazon baseline model	SASRec, NDCG@10 = 0.046315
Baseline cross-dataset rank agreement	Spearman $\rho = 0.939$
MovieLens nominal 10% recent median change	−6.73%
Amazon nominal 10% recent median change	−63.54%
MovieLens maximum unseen test targets	2.320%, early-profile 10%
Amazon maximum unseen test targets	13.177%, early-profile 10%
Most frequent condition winner	SASRec, 17 of 20 dataset–condition combinations
Other condition winner	GRU4Rec, 3 of 20 combinations
Severe random robustness agreement	$\rho = 0.467$
Severe recent robustness agreement	$\rho = -0.152$
Severe early robustness agreement	$\rho = 0.309$

Table 38: Experimental conditions won by GRU4Rec rather than SASRec.

Dataset	Condition	Winning NDCG@10
MovieLens-1M	Random 25%	0.055318
MovieLens-1M	Recent 25%	0.074018
Amazon Video Games	Random 10%	0.020602

16 Reproducibility Records

Table 39: Provenance information recorded for final experimental runs.

Category	Recorded information
Source control	Git commit hash
Dataset identity	Interaction-file SHA256 hash
Catalogue identity	Item-catalogue SHA256 hash
Configuration identity	Configuration SHA256 hashes
Randomness	Model seed and sparsity seed
Software environment	Python, RecBole and PyTorch versions
Hardware/runtime	Device, GPU information and execution runtimes
Training output	Checkpoint path and validation score
Sequence configuration	Effective maximum item-list length
BERT4Rec compatibility	Patch-active status
Evaluation configuration	Time field, metric and experimental condition metadata

A Detailed Experimental Results

This appendix reports the complete test-metric results supporting the main analysis. The experimental protocol evaluates Recall, HitRate, NDCG, and MRR at $K \in \{5, 10, 20\}$. Because the temporal leave-one-out evaluation contains exactly one relevant test target per user, HitRate@K is numerically identical to Recall@K in every experimental condition. HitRate is therefore not duplicated in the tables below.

Recall@5, Recall@10, Recall@20, NDCG@5, NDCG@10, NDCG@20, MRR@5, MRR@10, and MRR@20 are all reported. NDCG@10 is used as the primary metric for detailed robustness and ranking analysis. This is also the validation metric used for model selection.

For stochastic models, table entries are means across seeds 2025, 2026, and 2027. Deterministic models use their single final run.

B Complete Test-Metric Tables

In the following tables, R denotes Recall and N denotes NDCG.

B.1 MovieLens-1M

Table 40: Complete MovieLens-1M test metrics across all experimental conditions. Values are mean test scores across seeds for stochastic models and the single test score for deterministic models.

Scenario	Ret.	Model	R@5	R@10	R@20	N@5	N@10	N@20	MRR@5	MRR@10	MRR@20
Baseline	100%	SASRec	0.145067	0.223898	0.329356	0.093748	0.119079	0.145713	0.076947	0.087307	0.094604
Baseline	100%	GRU4Rec	0.134736	0.210032	0.308309	0.088337	0.112553	0.137308	0.073127	0.083042	0.089789
Baseline	100%	BERT4Rec	0.128328	0.212463	0.320959	0.079638	0.106708	0.133973	0.063744	0.074836	0.082232
Baseline	100%	EASE	0.051210	0.090322	0.153795	0.032601	0.045156	0.061191	0.026563	0.031692	0.036086
Baseline	100%	MultiVAE	0.043421	0.081151	0.139598	0.027150	0.039195	0.053790	0.021861	0.026741	0.030651
Baseline	100%	LightGCN	0.044084	0.079604	0.137609	0.026700	0.038068	0.052614	0.021059	0.025681	0.029610
Baseline	100%	BPR	0.042703	0.079218	0.138272	0.026360	0.038047	0.052876	0.021034	0.025787	0.029803
Baseline	100%	ItemKNN	0.042095	0.075572	0.130096	0.026617	0.037286	0.050919	0.021592	0.025903	0.029564
Baseline	100%	NeuMF	0.036681	0.067120	0.120816	0.022227	0.031950	0.045409	0.017505	0.021448	0.025081
Baseline	100%	Pop	0.021876	0.039443	0.075240	0.013566	0.019158	0.028064	0.010872	0.013127	0.015495
Random	50%	SASRec	0.101923	0.169594	0.260192	0.061537	0.083323	0.106119	0.048377	0.057310	0.063510
Random	50%	GRU4Rec	0.092145	0.154790	0.243840	0.057619	0.077763	0.100123	0.046378	0.054625	0.060679
Random	50%	BERT4Rec	0.097834	0.162579	0.259088	0.060532	0.081353	0.105620	0.048376	0.056901	0.063493
Random	50%	EASE	0.025522	0.046072	0.086841	0.015691	0.022259	0.032497	0.012504	0.015169	0.017943
Random	50%	MultiVAE	0.025301	0.047951	0.088278	0.014733	0.021925	0.032006	0.011308	0.014200	0.016907
Random	50%	LightGCN	0.023312	0.041819	0.078831	0.014052	0.019990	0.029202	0.011048	0.013471	0.015922
Random	50%	BPR	0.021600	0.038780	0.073252	0.012868	0.018340	0.026943	0.010042	0.012252	0.014555
Random	50%	ItemKNN	0.025853	0.045244	0.084355	0.016011	0.022253	0.032060	0.012825	0.015384	0.018034
Random	50%	NeuMF	0.021876	0.039609	0.077781	0.013224	0.018868	0.028392	0.010424	0.012701	0.015248
Random	50%	Pop	0.017236	0.029831	0.057673	0.011922	0.015888	0.022835	0.010176	0.011751	0.013607
Random	25%	SASRec	0.067065	0.118384	0.186941	0.035886	0.052454	0.069682	0.025734	0.032556	0.037229
Random	25%	GRU4Rec	0.064192	0.111369	0.177273	0.040165	0.055318	0.071905	0.032357	0.038549	0.043062
Random	25%	BERT4Rec	0.055187	0.093526	0.157772	0.034110	0.046387	0.062492	0.027235	0.032232	0.036579
Random	25%	EASE	0.019059	0.037454	0.070434	0.010945	0.016818	0.025051	0.008339	0.010717	0.012922
Random	25%	MultiVAE	0.019556	0.038062	0.069164	0.011124	0.016999	0.024754	0.008383	0.010744	0.012817
Random	25%	LightGCN	0.016296	0.032206	0.061540	0.010003	0.015103	0.022439	0.007960	0.010038	0.012009
Random	25%	BPR	0.015247	0.031543	0.056955	0.008992	0.014214	0.020566	0.006952	0.009079	0.010785
Random	25%	ItemKNN	0.014584	0.028339	0.055684	0.008565	0.012899	0.019746	0.006632	0.008353	0.010198
Random	25%	NeuMF	0.016849	0.030549	0.061043	0.010143	0.014506	0.022139	0.007967	0.009729	0.011784
Random	25%	Pop	0.012264	0.025191	0.054027	0.007248	0.011369	0.018559	0.005599	0.007266	0.009187
Random	10%	SASRec	0.033864	0.063474	0.108717	0.017052	0.026539	0.037860	0.011573	0.015437	0.018484
Random	10%	GRU4Rec	0.025798	0.047785	0.086952	0.015859	0.022897	0.032695	0.012636	0.015497	0.018132
Random	10%	BERT4Rec	0.019501	0.036294	0.065794	0.012017	0.017363	0.024747	0.009574	0.011732	0.013720
Random	10%	EASE	0.011932	0.025025	0.045244	0.007037	0.011213	0.016290	0.005458	0.007148	0.008523
Random	10%	MultiVAE	0.011988	0.024804	0.050989	0.007147	0.011223	0.017777	0.005576	0.007218	0.008981
Random	10%	LightGCN	0.011048	0.023533	0.049829	0.007190	0.011183	0.017696	0.005947	0.007569	0.009287
Random	10%	BPR	0.010827	0.022097	0.044691	0.006063	0.009634	0.015281	0.004516	0.005948	0.007464
Random	10%	ItemKNN	0.001326	0.002320	0.006132	0.000634	0.000957	0.001888	0.000414	0.000549	0.000787
Random	10%	NeuMF	0.013590	0.026240	0.049773	0.008034	0.012062	0.017957	0.006229	0.007855	0.009444
Random	10%	Pop	0.011601	0.030494	0.055353	0.007321	0.013303	0.019523	0.005933	0.008330	0.010000
Recent	50%	SASRec	0.116065	0.188211	0.286322	0.072833	0.095992	0.120670	0.058739	0.068197	0.074904
Recent	50%	GRU4Rec	0.107226	0.179428	0.273174	0.067441	0.090717	0.114222	0.054472	0.064035	0.070384
Recent	50%	BERT4Rec	0.100431	0.174235	0.269528	0.061408	0.085193	0.109181	0.048698	0.058468	0.064996
Recent	50%	EASE	0.047067	0.082864	0.135731	0.028556	0.040019	0.053297	0.022561	0.027225	0.030822
Recent	50%	MultiVAE	0.041101	0.075627	0.135344	0.024598	0.035683	0.050664	0.019240	0.023770	0.027821
Recent	50%	LightGCN	0.036681	0.067562	0.121534	0.022304	0.032217	0.045745	0.017641	0.021690	0.025340
Recent	50%	BPR	0.036570	0.067065	0.117335	0.022653	0.032380	0.045003	0.018123	0.022059	0.025478
Recent	50%	ItemKNN	0.039775	0.069108	0.120484	0.024566	0.033934	0.046878	0.019617	0.023414	0.026942
Recent	50%	NeuMF	0.036294	0.066402	0.117280	0.022222	0.031859	0.044569	0.017630	0.021548	0.024957
Recent	50%	Pop	0.018893	0.028505	0.054359	0.012682	0.015759	0.022232	0.010667	0.011919	0.013661
Recent	25%	SASRec	0.086841	0.153077	0.241686	0.051146	0.072425	0.094698	0.039499	0.048201	0.054245
Recent	25%	GRU4Rec	0.087891	0.146614	0.231798	0.055110	0.074018	0.095424	0.044414	0.052169	0.057976
Recent	25%	BERT4Rec	0.074964	0.130096	0.207491	0.045343	0.063069	0.082552	0.035692	0.042947	0.048251
Recent	25%	EASE	0.050713	0.088664	0.145177	0.031595	0.043787	0.057953	0.025384	0.030369	0.034189
Recent	25%	MultiVAE	0.047509	0.083913	0.139874	0.029293	0.040953	0.054982	0.023359	0.028107	0.031893
Recent	25%	LightGCN	0.041929	0.076787	0.132637	0.025887	0.037051	0.051074	0.020643	0.025188	0.028986
Recent	25%	BPR	0.044249	0.077837	0.132361	0.026993	0.037762	0.051426	0.021391	0.025781	0.029468

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Table 40 – continued from previous page

Scenario	Ret.	Model	R@5	R@10	R@20	N@5	N@10	N@20	MRR@5	MRR@10	MRR@20
Recent	25%	ItemKNN	0.046569	0.077892	0.127113	0.029064	0.039087	0.051417	0.023368	0.027443	0.030768
Recent	25%	NeuMF	0.041984	0.074135	0.122473	0.025878	0.036131	0.048264	0.020643	0.024791	0.028074
Recent	25%	Pop	0.018230	0.033643	0.057507	0.011788	0.016708	0.022801	0.009736	0.011728	0.013432
Recent	10%	SASRec	0.072147	0.125842	0.199260	0.037449	0.054733	0.073206	0.026087	0.033173	0.038197
Recent	10%	GRU4Rec	0.062811	0.102806	0.164457	0.038671	0.051480	0.066961	0.030813	0.036025	0.040218
Recent	10%	BERT4Rec	0.045796	0.077561	0.132693	0.028646	0.038849	0.052752	0.023055	0.027225	0.031023
Recent	10%	EASE	0.055022	0.089659	0.137057	0.034949	0.046078	0.058020	0.028397	0.032948	0.036204
Recent	10%	MultiVAE	0.048116	0.084908	0.135123	0.029670	0.041512	0.054142	0.023665	0.028520	0.031953
Recent	10%	LightGCN	0.049221	0.084521	0.138769	0.030236	0.041545	0.055199	0.024053	0.028658	0.032373
Recent	10%	BPR	0.042924	0.073583	0.118550	0.026928	0.036775	0.048035	0.021693	0.025718	0.028752
Recent	10%	ItemKNN	0.035797	0.054856	0.085018	0.023250	0.029457	0.036976	0.019142	0.021727	0.023735
Recent	10%	NeuMF	0.041432	0.069495	0.112143	0.025676	0.034686	0.045435	0.020541	0.024223	0.027156
Recent	10%	Pop	0.023699	0.035631	0.055187	0.013419	0.017196	0.022113	0.010073	0.011586	0.012922
Early	50%	SASRec	0.056126	0.092586	0.151641	0.034622	0.046350	0.061180	0.027624	0.032428	0.036447
Early	50%	GRU4Rec	0.028008	0.049995	0.087228	0.017131	0.024181	0.033487	0.013594	0.016468	0.018965
Early	50%	BERT4Rec	0.034361	0.064136	0.110871	0.020874	0.030360	0.042081	0.016487	0.020319	0.023487
Early	50%	EASE	0.024693	0.043255	0.076069	0.015275	0.021184	0.029363	0.012220	0.014604	0.016787
Early	50%	MultiVAE	0.021489	0.041045	0.073473	0.013615	0.019838	0.027961	0.011051	0.013560	0.015749
Early	50%	LightGCN	0.022097	0.040437	0.074467	0.013606	0.019482	0.027979	0.010837	0.013231	0.015508
Early	50%	BPR	0.021711	0.040769	0.070765	0.013664	0.019748	0.027248	0.011050	0.013516	0.015529
Early	50%	ItemKNN	0.022539	0.037454	0.067948	0.014497	0.019261	0.026868	0.011869	0.013800	0.015834
Early	50%	NeuMF	0.022760	0.041597	0.073638	0.013766	0.019758	0.027787	0.010842	0.013258	0.015424
Early	50%	Pop	0.014915	0.028671	0.056679	0.008571	0.012923	0.019964	0.006552	0.008293	0.010204
Early	25%	SASRec	0.038062	0.068280	0.111148	0.022574	0.032298	0.043075	0.017535	0.021520	0.024446
Early	25%	GRU4Rec	0.014363	0.026737	0.055132	0.008684	0.012610	0.019698	0.006838	0.008415	0.010312
Early	25%	BERT4Rec	0.014308	0.027345	0.051817	0.008584	0.012771	0.018903	0.006730	0.008443	0.010097
Early	25%	EASE	0.016407	0.029334	0.056182	0.010058	0.014183	0.020952	0.008007	0.009678	0.011526
Early	25%	MultiVAE	0.015689	0.028560	0.056899	0.009902	0.014025	0.021085	0.008020	0.009698	0.011581
Early	25%	LightGCN	0.013921	0.025688	0.057784	0.009105	0.012845	0.020852	0.007532	0.009039	0.011179
Early	25%	BPR	0.015634	0.028505	0.058668	0.009813	0.013881	0.021388	0.007923	0.009546	0.011544
Early	25%	ItemKNN	0.014915	0.027676	0.049221	0.009320	0.013377	0.018787	0.007505	0.009138	0.010603
Early	25%	NeuMF	0.015634	0.029444	0.058502	0.009525	0.013909	0.021182	0.007544	0.009306	0.011263
Early	25%	Pop	0.011767	0.021876	0.049553	0.006780	0.009965	0.016844	0.005151	0.006416	0.008241
Early	10%	SASRec	0.023643	0.045354	0.075351	0.014328	0.021299	0.028858	0.011300	0.014148	0.016210
Early	10%	GRU4Rec	0.015081	0.029444	0.054856	0.009146	0.013747	0.020062	0.007224	0.009098	0.010774
Early	10%	BERT4Rec	0.014584	0.029002	0.053806	0.008341	0.012949	0.019167	0.006337	0.008207	0.009886
Early	10%	EASE	0.013092	0.024362	0.046901	0.007941	0.011551	0.017149	0.006278	0.007748	0.009232
Early	10%	MultiVAE	0.013645	0.025798	0.047895	0.008614	0.012501	0.018027	0.006972	0.008552	0.010038
Early	10%	LightGCN	0.010772	0.019998	0.043421	0.006619	0.009538	0.015356	0.005268	0.006436	0.007978
Early	10%	BPR	0.012319	0.023367	0.049331	0.007478	0.011007	0.017457	0.005905	0.007335	0.009046
Early	10%	ItemKNN	0.010772	0.016904	0.032317	0.006771	0.008786	0.012631	0.005461	0.006312	0.007338
Early	10%	NeuMF	0.012098	0.023146	0.047398	0.007531	0.011035	0.017084	0.006039	0.007446	0.009063
Early	10%	Pop	0.014584	0.024196	0.050878	0.008980	0.012014	0.018613	0.007132	0.008341	0.010074

B.2 Amazon Video Games

Table 41: Complete Amazon Video Games test metrics across all experimental conditions. Values are mean test scores across seeds for stochastic models and the single test score for deterministic models.

Scenario	Ret.	Model	R@5	R@10	R@20	N@5	N@10	N@20	MRR@5	MRR@10	MRR@20
Baseline	100%	SASRec	0.060823	0.096300	0.142331	0.034908	0.046315	0.057919	0.026411	0.031080	0.034247
Baseline	100%	GRU4Rec	0.053916	0.084983	0.128507	0.034953	0.044938	0.055887	0.028756	0.032841	0.035818
Baseline	100%	BERT4Rec	0.047083	0.074291	0.112916	0.030387	0.039125	0.048845	0.024931	0.028502	0.031146
Baseline	100%	EASE	0.048365	0.072666	0.102613	0.032719	0.040555	0.048094	0.027594	0.030816	0.032868
Baseline	100%	MultiVAE	0.041353	0.064361	0.094528	0.027229	0.034618	0.042207	0.022608	0.025627	0.027691
Baseline	100%	LightGCN	0.042520	0.065933	0.098161	0.028336	0.035857	0.043960	0.023686	0.026761	0.028961
Baseline	100%	BPR	0.042278	0.065002	0.096221	0.028151	0.035450	0.043304	0.023520	0.026504	0.028640
Baseline	100%	ItemKNN	0.028748	0.043934	0.063835	0.019178	0.024051	0.029061	0.016035	0.018025	0.019387
Baseline	100%	NeuMF	0.029505	0.046247	0.070559	0.018951	0.024331	0.030441	0.015506	0.017706	0.019364
Baseline	100%	Pop	0.017772	0.032012	0.046678	0.011771	0.016255	0.019940	0.009814	0.011597	0.012597
Random	50%	SASRec	0.043777	0.070637	0.106308	0.024888	0.033523	0.042499	0.018690	0.022224	0.024665
Random	50%	GRU4Rec	0.034047	0.054684	0.084220	0.021694	0.028327	0.035752	0.017658	0.020371	0.022387
Random	50%	BERT4Rec	0.026593	0.043198	0.067730	0.016957	0.022285	0.028450	0.013807	0.015981	0.017654
Random	50%	EASE	0.030467	0.045148	0.063646	0.020500	0.025237	0.029889	0.017226	0.019174	0.020437
Random	50%	MultiVAE	0.027581	0.043918	0.067368	0.017853	0.023091	0.028984	0.014679	0.016814	0.018413
Random	50%	LightGCN	0.031419	0.048497	0.073381	0.020894	0.026374	0.032628	0.017450	0.019688	0.021384
Random	50%	BPR	0.028328	0.044197	0.066958	0.018389	0.023496	0.029224	0.015141	0.017233	0.018792
Random	50%	ItemKNN	0.007317	0.012095	0.019602	0.004886	0.006410	0.008281	0.004091	0.004708	0.005207
Random	50%	NeuMF	0.019349	0.031565	0.048570	0.012375	0.016301	0.020587	0.010098	0.011704	0.012872
Random	50%	Pop	0.016038	0.031965	0.045905	0.010828	0.015887	0.019393	0.009130	0.011164	0.012118
Random	25%	SASRec	0.031823	0.050500	0.077781	0.018971	0.024978	0.031843	0.014770	0.017229	0.019097
Random	25%	GRU4Rec	0.026619	0.042972	0.065865	0.016932	0.022204	0.027957	0.013775	0.015941	0.017502
Random	25%	BERT4Rec	0.015696	0.026934	0.041096	0.010710	0.014335	0.017896	0.009085	0.010576	0.011543
Random	25%	EASE	0.012363	0.018845	0.028543	0.008330	0.010419	0.012852	0.007012	0.007870	0.008528
Random	25%	MultiVAE	0.016253	0.027434	0.042820	0.010357	0.013955	0.017829	0.008431	0.009906	0.010961
Random	25%	LightGCN	0.015622	0.024490	0.036922	0.010176	0.013017	0.016133	0.008395	0.009552	0.010393
Random	25%	BPR	0.007974	0.012626	0.020448	0.005108	0.006598	0.008560	0.004173	0.004780	0.005310
Random	25%	ItemKNN	0.001719	0.003659	0.007538	0.001034	0.001659	0.002635	0.000812	0.001068	0.001333
Random	25%	NeuMF	0.018167	0.030756	0.046357	0.011812	0.015859	0.019779	0.009752	0.011406	0.012470
Random	25%	Pop	0.016558	0.030215	0.045448	0.010994	0.015364	0.019241	0.009196	0.010973	0.012053
Random	10%	SASRec	0.024585	0.040323	0.061018	0.014942	0.020016	0.025217	0.011793	0.013878	0.015289
Random	10%	GRU4Rec	0.024464	0.039482	0.059651	0.015768	0.020602	0.025664	0.012930	0.014913	0.016282
Random	10%	BERT4Rec	0.017946	0.028154	0.043104	0.011533	0.014880	0.018625	0.009440	0.010849	0.011858
Random	10%	EASE	0.004400	0.008074	0.013751	0.002957	0.004134	0.005552	0.002487	0.002966	0.003345
Random	10%	MultiVAE	0.013867	0.024559	0.040002	0.008602	0.012034	0.015900	0.006891	0.008293	0.009335
Random	10%	LightGCN	0.008090	0.013083	0.020590	0.005265	0.006863	0.008748	0.004340	0.004990	0.005501
Random	10%	BPR	0.005293	0.008816	0.014629	0.003310	0.004441	0.005896	0.002665	0.003126	0.003518
Random	10%	ItemKNN	0.002839	0.004510	0.008027	0.001700	0.002242	0.003127	0.001318	0.001543	0.001783
Random	10%	NeuMF	0.016994	0.030131	0.043509	0.011105	0.015325	0.018726	0.009188	0.010914	0.011859
Random	10%	Pop	0.017315	0.030483	0.047072	0.011425	0.015739	0.019893	0.009509	0.011322	0.012443
Recent	50%	SASRec	0.050289	0.079500	0.117715	0.027290	0.036706	0.046328	0.019714	0.023583	0.026203
Recent	50%	GRU4Rec	0.037542	0.060576	0.094875	0.023642	0.031041	0.039660	0.019107	0.022133	0.024470
Recent	50%	BERT4Rec	0.029295	0.047204	0.073213	0.018645	0.024402	0.030931	0.015174	0.017530	0.019297
Recent	50%	EASE	0.039676	0.057701	0.080614	0.026726	0.032524	0.038308	0.022470	0.024845	0.026428
Recent	50%	MultiVAE	0.035534	0.055310	0.082013	0.023159	0.029517	0.036236	0.019112	0.021714	0.023541
Recent	50%	LightGCN	0.039403	0.059436	0.087900	0.026273	0.032723	0.039882	0.021973	0.024620	0.026564
Recent	50%	BPR	0.036832	0.056166	0.083511	0.024409	0.030631	0.037502	0.020341	0.022891	0.024756
Recent	50%	ItemKNN	0.013515	0.021021	0.032296	0.008908	0.011309	0.014134	0.007400	0.008376	0.009137
Recent	50%	NeuMF	0.022514	0.035592	0.053848	0.014621	0.018819	0.023411	0.012038	0.013752	0.015000
Recent	50%	Pop	0.018577	0.031918	0.048082	0.012136	0.016376	0.020449	0.010037	0.011743	0.012854
Recent	25%	SASRec	0.041295	0.066028	0.096989	0.023475	0.031443	0.039232	0.017625	0.020896	0.023013
Recent	25%	GRU4Rec	0.027833	0.046489	0.072593	0.017329	0.023328	0.029886	0.013908	0.016365	0.018142
Recent	25%	BERT4Rec	0.016411	0.028123	0.043440	0.010377	0.014126	0.017977	0.008413	0.009939	0.010984
Recent	25%	EASE	0.022866	0.033353	0.047167	0.015670	0.019045	0.022520	0.013305	0.014688	0.015632
Recent	25%	MultiVAE	0.023823	0.037726	0.057543	0.015477	0.019948	0.024934	0.012747	0.014578	0.015934
Recent	25%	LightGCN	0.025147	0.038399	0.057281	0.016439	0.020691	0.025431	0.013592	0.015327	0.016610
Recent	25%	BPR	0.017184	0.026372	0.040144	0.011296	0.014238	0.017701	0.009366	0.010564	0.011504
Recent	25%	ItemKNN	0.005740	0.009982	0.018466	0.003770	0.005126	0.007257	0.003132	0.003683	0.004260
Recent	25%	NeuMF	0.018892	0.031513	0.048034	0.012075	0.016118	0.020279	0.009855	0.011501	0.012635
Recent	25%	Pop	0.019428	0.031318	0.049438	0.012462	0.016221	0.020793	0.010207	0.011709	0.012961
Recent	10%	SASRec	0.037868	0.060440	0.088694	0.020503	0.027772	0.034894	0.014780	0.017763	0.019706
Recent	10%	GRU4Rec	0.025741	0.041306	0.062474	0.016243	0.021246	0.026584	0.013141	0.015187	0.016645
Recent	10%	BERT4Rec	0.016957	0.028217	0.046552	0.010877	0.014456	0.019082	0.008883	0.010324	0.011588

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Table 41 – continued from previous page

Scenario	Ret.	Model	R@5	R@10	R@20	N@5	N@10	N@20	MRR@5	MRR@10	MRR@20
Recent	10%	EASE	0.009903	0.016243	0.024821	0.006666	0.008705	0.010851	0.005605	0.006440	0.007017
Recent	10%	MultiVAE	0.014497	0.025132	0.039866	0.009063	0.012462	0.016161	0.007288	0.008668	0.009670
Recent	10%	LightGCN	0.009520	0.014876	0.023192	0.006247	0.007964	0.010051	0.005180	0.005879	0.006444
Recent	10%	BPR	0.005672	0.009483	0.016217	0.003571	0.004792	0.006485	0.002887	0.003384	0.003843
Recent	10%	ItemKNN	0.005031	0.008153	0.013940	0.003079	0.004084	0.005534	0.002431	0.002843	0.003234
Recent	10%	NeuMF	0.018135	0.030152	0.046284	0.011887	0.015770	0.019842	0.009854	0.011454	0.012569
Recent	10%	Pop	0.017788	0.032139	0.048712	0.011730	0.016263	0.020420	0.009759	0.011565	0.012689
Early	50%	SASRec	0.044817	0.070490	0.104421	0.025729	0.033990	0.042541	0.019451	0.022837	0.025169
Early	50%	GRU4Rec	0.031645	0.050258	0.077408	0.020168	0.026140	0.032962	0.016417	0.018854	0.020704
Early	50%	BERT4Rec	0.024438	0.039671	0.061728	0.015460	0.020346	0.025887	0.012532	0.014525	0.016027
Early	50%	EASE	0.025105	0.036869	0.052891	0.016253	0.020019	0.024052	0.013354	0.014886	0.015984
Early	50%	MultiVAE	0.023244	0.037369	0.057223	0.014732	0.019265	0.024255	0.011953	0.013805	0.015158
Early	50%	LightGCN	0.027344	0.042525	0.063888	0.018042	0.022922	0.028296	0.015004	0.017001	0.018462
Early	50%	BPR	0.025284	0.039703	0.060482	0.016551	0.021172	0.026392	0.013693	0.015576	0.016992
Early	50%	ItemKNN	0.008106	0.012490	0.020154	0.005120	0.006516	0.008447	0.004143	0.004706	0.005234
Early	50%	NeuMF	0.019186	0.030351	0.046300	0.012281	0.015864	0.019876	0.010028	0.011491	0.012581
Early	50%	Pop	0.016148	0.025594	0.042830	0.009826	0.012926	0.017277	0.007728	0.009034	0.010225
Early	25%	SASRec	0.024080	0.038530	0.058001	0.015243	0.019885	0.024787	0.012360	0.014257	0.015592
Early	25%	GRU4Rec	0.021473	0.035014	0.052970	0.013506	0.017853	0.022375	0.010906	0.012681	0.013913
Early	25%	BERT4Rec	0.015260	0.023675	0.036097	0.009122	0.011838	0.014961	0.007128	0.008246	0.009093
Early	25%	EASE	0.011323	0.017630	0.026241	0.007593	0.009622	0.011787	0.006375	0.007206	0.007794
Early	25%	MultiVAE	0.013914	0.023686	0.039729	0.008777	0.011901	0.015898	0.007104	0.008373	0.009439
Early	25%	LightGCN	0.014676	0.023344	0.035682	0.009516	0.012300	0.015408	0.007830	0.008968	0.009814
Early	25%	BPR	0.008389	0.013315	0.021184	0.005351	0.006928	0.008900	0.004356	0.004999	0.005530
Early	25%	ItemKNN	0.002870	0.005109	0.009572	0.001829	0.002548	0.003668	0.001492	0.001787	0.002089
Early	25%	NeuMF	0.016916	0.027923	0.041106	0.010867	0.014422	0.017721	0.008899	0.010363	0.011251
Early	25%	Pop	0.015312	0.028322	0.039692	0.008554	0.012809	0.015700	0.006402	0.008184	0.008987
Early	10%	SASRec	0.019144	0.032680	0.046994	0.012366	0.016713	0.020306	0.010161	0.011938	0.012911
Early	10%	GRU4Rec	0.018508	0.032343	0.047809	0.012099	0.016547	0.020401	0.010010	0.011831	0.012860
Early	10%	BERT4Rec	0.015896	0.024406	0.036070	0.009355	0.012086	0.015023	0.007220	0.008336	0.009136
Early	10%	EASE	0.004179	0.007144	0.012553	0.002925	0.003878	0.005234	0.002517	0.002907	0.003273
Early	10%	MultiVAE	0.013289	0.023165	0.038152	0.007931	0.011106	0.014863	0.006200	0.007499	0.008513
Early	10%	LightGCN	0.007622	0.012137	0.018555	0.004966	0.006417	0.008025	0.004099	0.004692	0.005126
Early	10%	BPR	0.004947	0.008142	0.013494	0.003127	0.004154	0.005491	0.002535	0.002955	0.003313
Early	10%	ItemKNN	0.002334	0.004116	0.007002	0.001397	0.001965	0.002693	0.001094	0.001323	0.001522
Early	10%	NeuMF	0.015223	0.027355	0.038835	0.009205	0.013123	0.016002	0.007224	0.008838	0.009615
Early	10%	Pop	0.015659	0.027755	0.039692	0.010865	0.014770	0.017768	0.009298	0.010906	0.011718

C NDCG@10 Robustness

For the robustness tables, signed relative change is calculated as

$$\Delta = \frac{\text{NDCG@10}_{\text{sparse}} - \text{NDCG@10}_{\text{baseline}}}{\text{NDCG@10}_{\text{baseline}}} \times 100. \quad (5)$$

Negative values indicate degradation relative to the same model’s full-data baseline, while positive values indicate improvement. A smaller loss does not necessarily imply stronger absolute recommendation performance; relative change should therefore be interpreted together with the absolute NDCG@10 values above.

C.1 MovieLens-1M

Table 42: MovieLens-1M relative NDCG@10 change from each model’s full-data baseline. Negative values indicate degradation and positive values indicate improvement.

Model	Random per-user			Recent-history			Early-profile		
	50%	25%	10%	50%	25%	10%	50%	25%	10%
SASRec	-30.03%	-55.95%	-77.71%	-19.39%	-39.18%	-54.04%	-61.08%	-72.88%	-82.11%
GRU4Rec	-30.91%	-50.85%	-79.66%	-19.40%	-34.24%	-54.26%	-78.52%	-88.80%	-87.79%
BERT4Rec	-23.76%	-56.53%	-83.73%	-20.16%	-40.90%	-63.59%	-71.55%	-88.03%	-87.86%
EASE	-50.71%	-62.76%	-75.17%	-11.38%	-3.03%	+2.04%	-53.09%	-68.59%	-74.42%
MultiVAE	-44.06%	-56.63%	-71.37%	-8.96%	+4.48%	+5.91%	-49.39%	-64.22%	-68.11%
LightGCN	-47.49%	-60.33%	-70.62%	-15.37%	-2.67%	+9.13%	-48.82%	-66.26%	-74.95%
BPR	-51.80%	-62.64%	-74.68%	-14.89%	-0.75%	-3.34%	-48.09%	-63.52%	-71.07%
ItemKNN	-40.32%	-65.41%	-97.43%	-8.99%	+4.83%	-21.00%	-48.34%	-64.12%	-76.44%
NeuMF	-40.95%	-54.60%	-62.25%	-0.29%	+13.09%	+8.56%	-38.16%	-56.47%	-65.46%
Pop	-17.07%	-40.66%	-30.56%	-17.74%	-12.79%	-10.24%	-32.55%	-47.99%	-37.29%

C.2 Amazon Video Games

Table 43: Amazon Video Games relative NDCG@10 change from each model’s full-data baseline. Negative values indicate degradation and positive values indicate improvement.

Model	Random per-user			Recent-history			Early-profile		
	50%	25%	10%	50%	25%	10%	50%	25%	10%
SASRec	-27.62%	-46.07%	-56.78%	-20.75%	-32.11%	-40.04%	-26.61%	-57.07%	-63.91%
GRU4Rec	-36.96%	-50.59%	-54.15%	-30.92%	-48.09%	-52.72%	-41.83%	-60.27%	-63.18%
BERT4Rec	-43.04%	-63.36%	-61.97%	-37.63%	-63.89%	-63.05%	-48.00%	-69.74%	-69.11%
EASE	-37.77%	-74.31%	-89.81%	-19.80%	-53.04%	-78.54%	-50.64%	-76.27%	-90.44%
MultiVAE	-33.30%	-59.69%	-65.24%	-14.74%	-42.38%	-64.00%	-44.35%	-65.62%	-67.92%
LightGCN	-26.45%	-63.70%	-80.86%	-8.74%	-42.30%	-77.79%	-36.07%	-65.70%	-82.10%
BPR	-33.72%	-81.39%	-87.47%	-13.59%	-59.84%	-86.48%	-40.27%	-80.46%	-88.28%
ItemKNN	-73.35%	-93.10%	-90.68%	-52.98%	-78.69%	-83.02%	-72.91%	-89.41%	-91.83%
NeuMF	-33.00%	-34.82%	-37.01%	-22.65%	-33.75%	-35.19%	-34.80%	-40.73%	-46.06%
Pop	-2.26%	-5.48%	-3.17%	+0.74%	-0.21%	+0.05%	-20.48%	-21.20%	-9.14%

D NDCG@10 Model Rankings

D.1 MovieLens-1M

Table 44: MovieLens-1M NDCG@10 model ranks across all experimental conditions. Rank 1 denotes the highest mean NDCG@10.

Model	Base	Random per-user			Recent-history			Early-profile		
		50%	25%	10%	50%	25%	10%	50%	25%	10%
SASRec	1	1	2	1	1	2	1	1	1	1
GRU4Rec	2	3	1	2	2	1	2	3	9	2
BERT4Rec	3	2	3	3	3	3	6	2	8	3
EASE	4	4	5	7	4	4	3	4	2	6
MultiVAE	5	6	4	6	5	5	5	5	3	4
LightGCN	6	7	6	8	8	8	4	8	7	9
BPR	7	9	8	9	7	7	7	7	5	8
ItemKNN	8	5	9	10	6	6	9	9	6	10
NeuMF	9	8	7	5	9	9	8	6	4	7
Pop	10	10	10	4	10	10	10	10	10	5

D.2 Amazon Video Games

Table 45: Amazon Video Games NDCG@10 model ranks across all experimental conditions. Rank 1 denotes the highest mean NDCG@10.

Model	Base	Random per-user			Recent-history			Early-profile		
		50%	25%	10%	50%	25%	10%	50%	25%	10%
SASRec	1	1	1	2	1	1	1	1	1	1
GRU4Rec	2	2	2	1	4	2	2	2	2	2
BERT4Rec	4	7	5	5	7	9	5	5	7	5
EASE	3	4	8	9	3	5	7	6	8	9
MultiVAE	7	6	6	6	6	4	6	7	6	6
LightGCN	5	3	7	7	2	3	8	3	5	7
BPR	6	5	9	8	5	8	9	4	9	8
ItemKNN	9	10	10	10	10	10	10	10	10	10
NeuMF	8	8	3	4	8	7	4	8	3	4
Pop	10	9	4	3	9	6	3	9	4	3

E Aggregate Sparsity Effects

Table 46: Median signed relative NDCG@10 change across the ten models.

Scenario	Nominal retention	MovieLens	Amazon
Random	50%	-40.63%	-33.51%
Random	25%	-56.58%	-61.52%
Random	10%	-74.92%	-63.60%
Recent	50%	-15.13%	-20.27%
Recent	25%	-2.85%	-45.23%
Recent	10%	-6.79%	-63.53%
Early	50%	-49.11%	-41.05%
Early	25%	-65.24%	-65.66%
Early	10%	-74.68%	-68.51%

Table 47: Number of model-retention comparisons for which each sparsity mechanism produced the highest mean NDCG@10. There are 30 comparisons per dataset.

Best sparsity mechanism	MovieLens-1M	Amazon Video Games
Recent-history	29	28
Random per-user	1	2
Early-profile	0	0

The only MovieLens exception is Pop at 50% retention, where random per-user retention gives a slightly higher NDCG@10 than recent-history retention. The Amazon exceptions are BERT4Rec at 25% and 10% retention, where random per-user retention gives the highest NDCG@10 among the three sparsity mechanisms.

Table 48: Highest mean NDCG@10 model in each experimental condition.

Condition	MovieLens-1M	Amazon Video Games
Baseline	SASRec	SASRec
Random 50%	SASRec	SASRec
Random 25%	GRU4Rec	SASRec
Random 10%	SASRec	GRU4Rec
Recent 50%	SASRec	SASRec
Recent 25%	GRU4Rec	SASRec
Recent 10%	SASRec	SASRec
Early 50%	SASRec	SASRec
Early 25%	SASRec	SASRec
Early 10%	SASRec	SASRec

Across the 20 dataset-condition combinations, SASRec achieves the highest mean NDCG@10 in 17 conditions and GRU4Rec in the remaining three.

F Dataset Conditions Under Sparsity

Table 49: Nominal versus actual aggregate training retention.

Dataset	Nominal retention	Actual retention	Mean train interactions/user
MovieLens	100%	100.000%	93.19
MovieLens	50%	50.270%	46.85
MovieLens	25%	25.404%	23.67
MovieLens	10%	10.479%	9.76
Amazon	100%	100.000%	6.41
Amazon	50%	54.712%	3.51
Amazon	25%	29.767%	1.91
Amazon	10%	18.380%	1.18

Table 50: Percentage of held-out test targets whose items are globally absent from the retained training interactions.

Scenario	Retention	MovieLens	Amazon
Baseline	100%	0.000%	0.358%
Random	50%	0.033%	1.667%
Random	25%	0.265%	4.889%
Random	10%	1.425%	9.071%
Recent	50%	0.000%	1.116%
Recent	25%	0.066%	3.301%
Recent	10%	0.580%	6.894%
Early	50%	0.099%	3.331%
Early	25%	0.563%	8.621%
Early	10%	2.320%	13.177%

Table 51: Dataset characteristics under nominal 10% training retention.

Statistic	MovieLens-1M	Amazon Video Games
Actual aggregate retention	10.479%	18.380%
Mean retained train interactions/user	9.76	1.18
Random unseen test targets	1.425%	9.071%
Recent unseen test targets	0.580%	6.894%
Early unseen test targets	2.320%	13.177%

G Supporting Metric Consistency

Table 52: Spearman agreement between NDCG@10 model rankings and the supporting Recall@10 and MRR@10 rankings.

Condition	ML NDCG–Recall	ML NDCG–MRR	Amazon NDCG–Recall	Amazon NDCG–MRR
Baseline	0.988	0.952	0.988	0.976
Random 50%	0.964	0.988	0.988	1.000
Random 25%	0.976	1.000	0.988	1.000
Random 10%	0.988	0.952	0.988	1.000
Recent 50%	0.988	1.000	0.964	0.939
Recent 25%	0.988	1.000	0.976	0.988
Recent 10%	0.988	0.988	1.000	1.000
Early 50%	0.988	0.867	0.988	0.988
Early 25%	0.927	0.976	0.952	0.927
Early 10%	0.976	0.964	1.000	1.000

Table 53: Spearman agreement of NDCG model rankings across recommendation-list lengths.

Condition	ML @5 vs @10	ML @10 vs @20	Amazon @5 vs @10	Amazon @10 vs @20
Baseline	0.988	0.988	0.976	0.988
Random 50%	0.988	1.000	1.000	1.000
Random 25%	0.988	0.988	1.000	1.000
Random 10%	0.952	0.988	0.964	1.000
Recent 50%	1.000	0.988	0.976	0.988
Recent 25%	1.000	0.988	0.988	0.988
Recent 10%	0.988	1.000	0.988	1.000
Early 50%	0.842	0.927	0.988	0.988
Early 25%	0.976	0.794	0.915	0.964
Early 10%	0.939	0.952	0.988	0.988

These correlations show that the principal NDCG@10 ranking conclusions are broadly consistent with Recall@10 and MRR@10, and are generally stable across $K = 5, 10, 20$.

H Cross-Dataset Agreement

Table 54: Spearman agreement between MovieLens and Amazon NDCG@10 model rankings under matching conditions.

Condition	Spearman ρ
Baseline	0.939
Random 50%	0.491
Random 25%	0.552
Random 10%	0.915
Recent 50%	0.515
Recent 25%	0.382
Recent 10%	0.370
Early 50%	0.648
Early 25%	-0.018
Early 10%	0.794

Table 55: Cross-dataset agreement in model relative NDCG@10 change at nominal 10% retention.

Scenario	Spearman ρ
Random per-user	0.467
Recent-history	-0.152
Early-profile	0.309

I Seed Variability Under Sparse Conditions

I.1 MovieLens-1M

Table 56: Largest MovieLens sparse-condition NDCG@10 coefficients of variation among stochastic models.

Scenario	Retention	Model	SD	CV
Early-profile	10%	SASRec	0.009363	43.96%
Early-profile	10%	BERT4Rec	0.002247	17.35%
Early-profile	50%	GRU4Rec	0.003654	15.11%
Random per-user	10%	NeuMF	0.001589	13.18%
Random per-user	25%	LightGCN	0.001894	12.54%
Early-profile	25%	NeuMF	0.001554	11.17%
Random per-user	25%	BPR	0.001337	9.40%
Recent-history	25%	NeuMF	0.003340	9.24%

I.2 Amazon Video Games

Table 57: Largest Amazon sparse-condition NDCG@10 coefficients of variation among stochastic models.

Scenario	Retention	Model	SD	CV
Early-profile	25%	BPR	0.001631	23.54%
Early-profile	10%	BERT4Rec	0.002325	19.23%
Early-profile	25%	SASRec	0.003371	16.95%
Recent-history	10%	MultiVAE	0.001768	14.18%
Early-profile	25%	MultiVAE	0.001465	12.31%
Random per-user	10%	MultiVAE	0.001380	11.47%
Random per-user	25%	MultiVAE	0.001044	7.48%
Early-profile	25%	BERT4Rec	0.000876	7.40%

J Reporting Conventions

Table 58: Reporting conventions used for the experimental results.

Issue	Reporting convention
Primary metric	NDCG@10 is used for the main robustness, ranking, and cross-dataset analysis.
Full metric reporting	Recall, NDCG, and MRR are reported at $K = 5, 10, 20$ in the complete result tables.
HitRate	HitRate@K was computed as specified in the protocol but is not duplicated because it is numerically identical to Recall@K with one held-out relevant target per user.
Relative robustness	Signed relative NDCG@10 change is reported relative to each model’s own baseline. Negative values indicate degradation; positive values indicate improvement.
Absolute versus relative performance	Relative robustness is always interpreted together with absolute NDCG@10 because a low-performing model may exhibit only a small percentage loss.
Random scenario	The repository label <code>global</code> is reported as random per-user retention because thinning is performed independently within each user’s training history.
Early-profile scenario	The condition is described as early-profile or cold-start-like rather than true cold start because users remain known and retain training interactions.
Unseen target	An unseen test target is a target item globally absent from the retained training set, not merely an item absent from that user’s retained history.
Statistical interpretation	Stochastic-model results are descriptive means across three seeds; statistical significance is not claimed.